



WSQ · INSTRUCTOR-LED · 2 DAYS

Developing AI Apps and Agents on Azure (AI-103)

Course Code: TGS-2023036651

Tertiary Infotech Academy Pte Ltd
UEN 201200696W

Version 5.0 · 27 Sep 2026



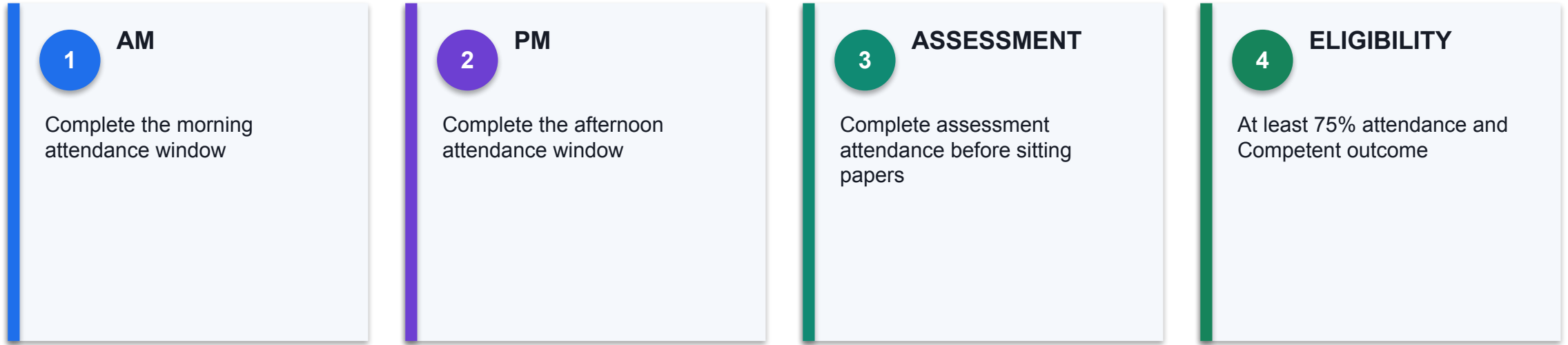
TOPIC 00

00

Course administration

WSQ requirements · learning outcomes · evidence expectations

Digital Attendance (Mandatory)



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

About the Trainer



Tertiary Infotech Academy Trainer

Microsoft Azure AI
training team

NAME	Tertiary Infotech Academy Trainer
TITLE / DESIGNATION	WSQ Technical Trainer
QUALIFICATIONS	Azure AI and adult-learning credentials
AREAS OF EXPERTISE	Microsoft Foundry, Azure AI services, MLOps
EXPERIENCE	Applied cloud AI delivery and facilitation
CONTACT	Introduced during the opening briefing

About the Trainer



Dr. Alfred Ang

Principal Trainer
AI, automation, and software engineering

NAME	Dr. Alfred Ang
TITLE / DESIGNATION	Principal Trainer
QUALIFICATIONS	PhD; Azure AI and software engineering
AREAS OF EXPERTISE	AI solutions, agents, automation, apps
EXPERIENCE	Industry implementation and adult learning
CONTACT	Shared during class

Ground Rules

1

PUNCTUAL

Return from breaks on time

2

PARTICIPATE

Explain decisions and evidence

3

PROTECT DATA

Use synthetic/non-confidential inputs

4

ASK EARLY

Raise blockers before they cascade

ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Learning Outcomes

1

LO1

Analyse Azure AI algorithms and efficiency

2

LO2

Evaluate strengths and limitations

3

LO3

Assess feasibility and improvements

4

EVIDENCE

Explain what proves each conclusion

ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Two-Day Lesson Plan

1

DAY 1 AM

Foundry project, model evaluation, chat app

2

DAY 1 PM

Agents, custom tools, vision

3

DAY 2 AM

Text analysis, text agent, content understanding

4

DAY 2 PM

Knowledge mining + 2-hour assessment

ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

AI-103 Course Scope

1

EXAM

AI-103: Developing AI Apps
and Agents on Azure

2

SCOPE

Skills measured from 16 Apr
2026

3

COURSE

WSQ TGS-2023036651 · two
days

4

FOCUS

Foundry apps, agents,
multimodal and extraction

ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Current Microsoft certification pathway

AI-103: Developing AI Apps and Agents on Azure · skills measured from 16 Apr 2026

[Microsoft Learn · AI-103 Study Guide](#)

OPEN RESOURCE

Briefing for Assessment

1

OPEN BOOK

Use approved slides and
Learner Guide

2

INDIVIDUAL

No discussion or shared
answers

3

EVIDENCE

Answer every K/A criterion

4

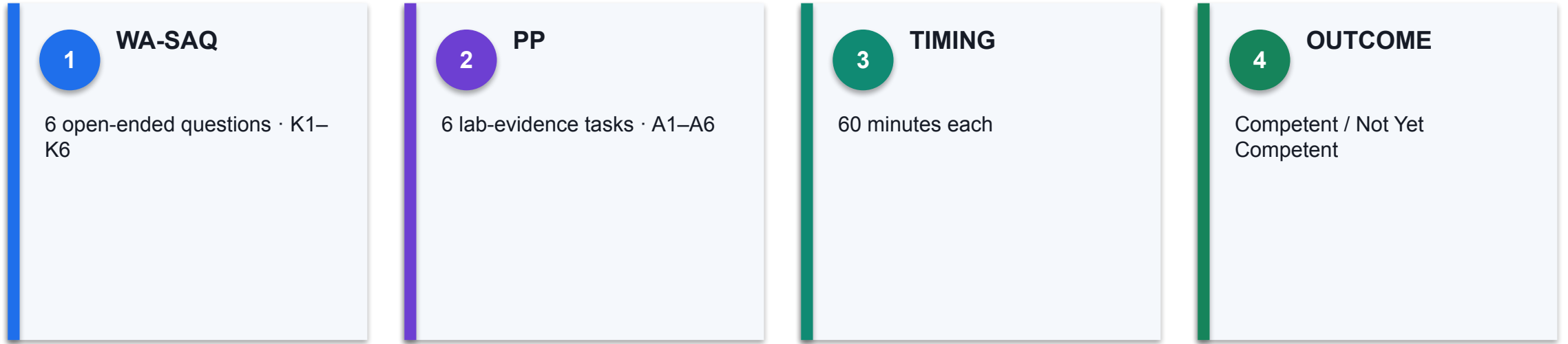
SUBMISSION

Upload through the LMS

ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

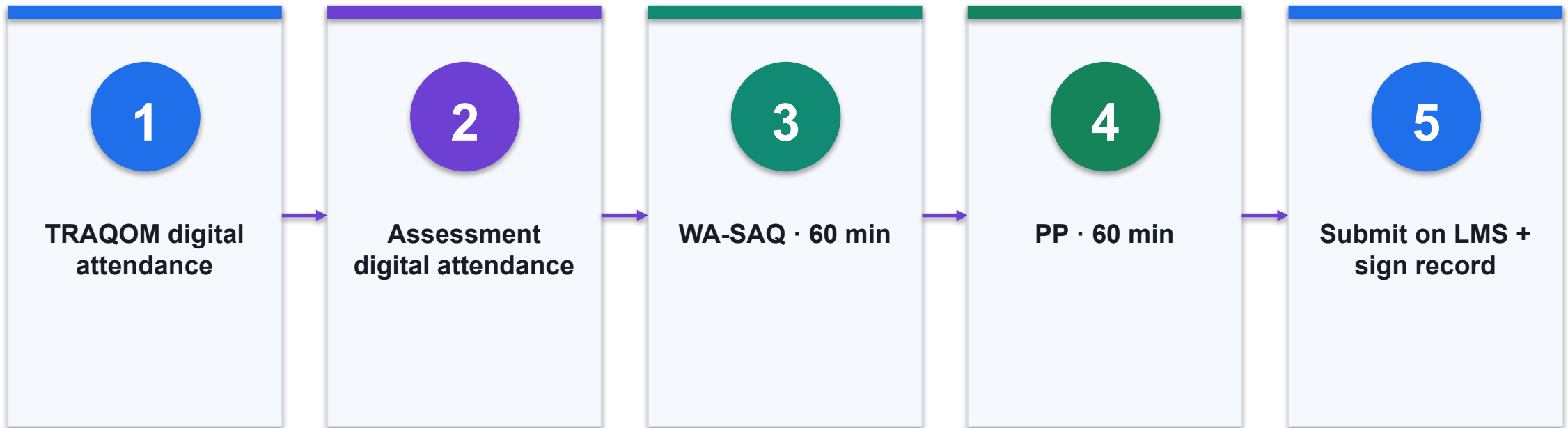
Assessment



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Assessment Flow



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Courseware and Assessment on the LMS

Download the learner materials and submit the two candidate papers on the LMS

[LMS-TMS · Courseware and Assessment](#)

OPEN RESOURCE

Access the Hands-On Labs

Open the ten numbered lab guides and the pinned, unchanged Microsoft Learning source snapshots

[GitHub · AI-103 Official Lab Materials](#)

OPEN RESOURCE

TOPIC 00

00

Day 1

Foundry planning · models · chat · agents · vision

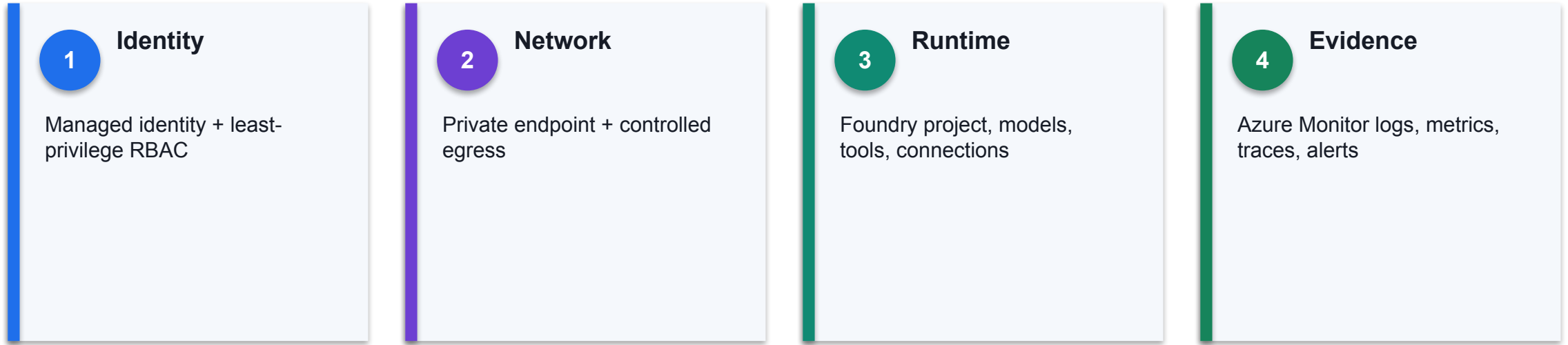
TOPIC 01

01

Plan a Foundry project

PLAN AND MANAGE · AI-103

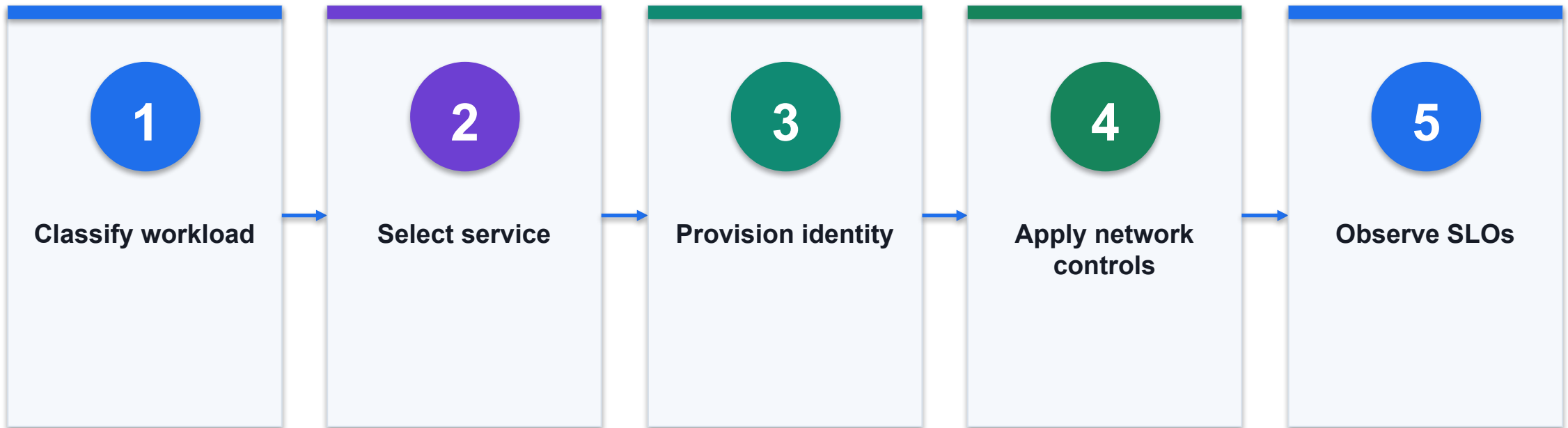
Plan a Foundry project · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Plan a Foundry project - runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Plan a Foundry project · exact contract

Contract	Exact value / field	Engineering consequence
Project endpoint	https://<resource>.services.ai.azure.com/api/projects/<project>	Foundry SDK
Auth	DefaultAzureCredential	Entra token; avoid embedded keys
RBAC	Foundry User	Project development; check endpoint-specific inference role
Diagnostics	RequestResponse + Audit	Send to Log Analytics

Plan a Foundry project · implementation excerpt

```
from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient
client = AIProjectClient(endpoint=PROJECT_ENDPOINT,

credential=DefaultAzureCredential())
for conn in client.connections.list():
    print(conn.name, conn.type)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

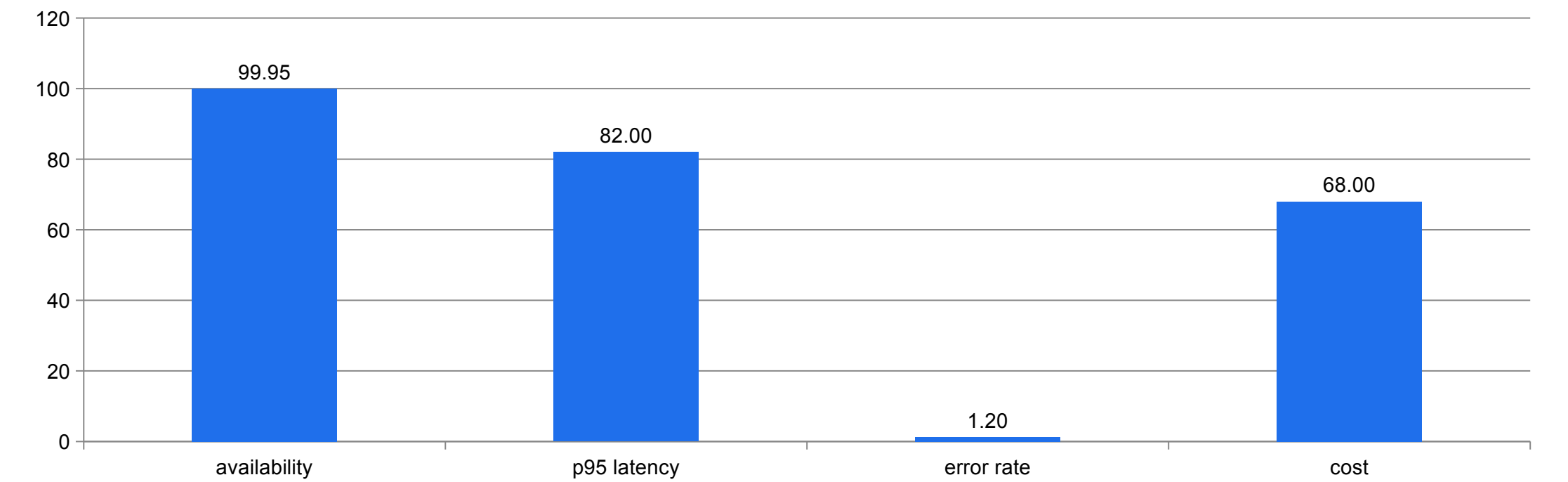
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

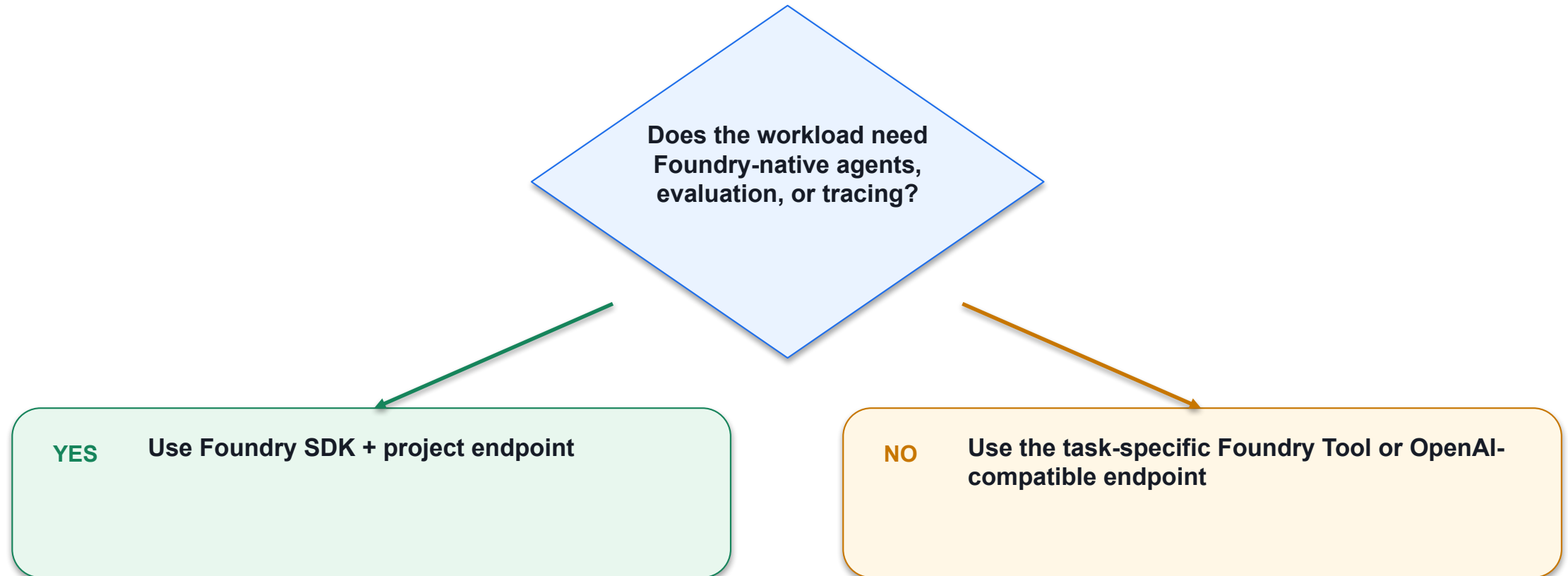
Handle auth, quota, timeout, and invalid-state paths.

Plan a Foundry project - engineering metric



WHAT THE DATA SHOWS Indicative dashboard: availability is a percentage; other values are normalized to target. Investigate any measure below its target band.

Plan a Foundry project · decision rule



Plan a Foundry project - failure and control

 401 / 403 LIKELY CAUSE Identity lacks data-plane role CONTROL / RECOVERY Verify principal, scope, and token audience	 429 LIKELY CAUSE Quota or rate limit CONTROL / RECOVERY Retry with backoff; request quota or scale units	 Timeout LIKELY CAUSE Network/DNS/private link CONTROL / RECOVERY Resolve FQDN from workload subnet	 Blind spot LIKELY CAUSE Diagnostics not routed CONTROL / RECOVERY Enable categories and alert ownership
---	---	---	--

A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 01: Plan a Foundry project

Mechanism focus: Plan a Foundry project



YOU WILL PRODUCE

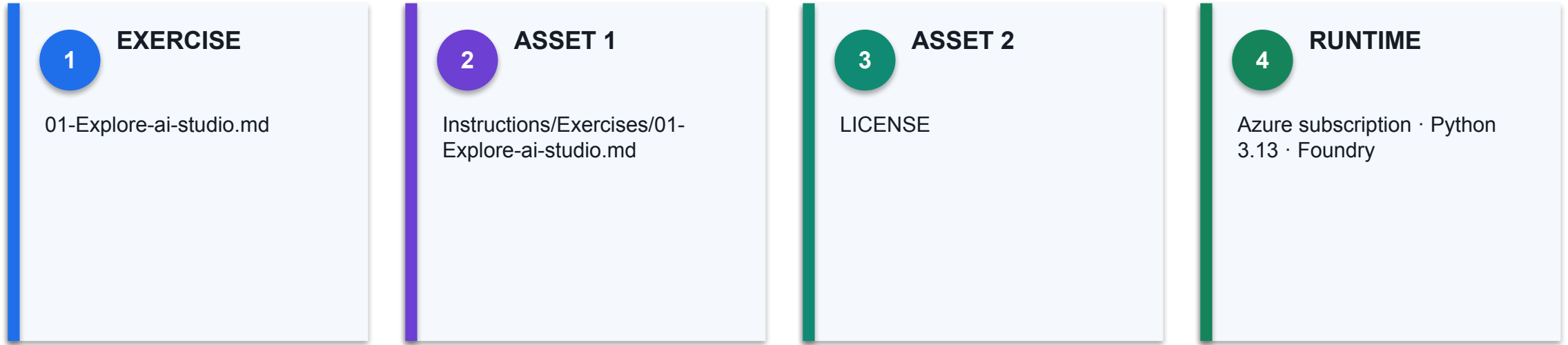
The official exercise completes with its expected output; a second learner can reproduce the result from the source instructions and the recorded environment.

Complete click-by-click procedure: Learner Guide · Lab 01 · [labs/lab-01-plan-a-foundry-project/README.md](#)

Official lab 01 · pinned source

Contract	Exact value / field	Engineering consequence
Repository	mslearn-ai-studio	MIT source snapshot
Revision	b702dc2fd9f9	SHA-256 per file in source manifest
Exercise	01-Explore-ai-studio.md	Unchanged instruction file
Domain	Plan and manage	AI-103 study guide

Official lab 01 · input boundaries



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Official lab 01 · implementation 1

```
from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient
client = AIProjectClient(endpoint=PROJECT_ENDPOINT,

credential=DefaultAzureCredential())
for conn in client.connections.list():
    print(conn.name, conn.type)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 01 · implementation 2

```
Project endpoint:  
https://<resource>.services.ai.azure.com/api/projects/<  
project> # Foundry SDK  
Auth: DefaultAzureCredential # Entra token; avoid  
embedded keys  
RBAC: Foundry User # Project development; check  
endpoint-specific inference role  
Diagnostics: RequestResponse + Audit # Send to Log  
Analytics
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 01 · implementation 3

Identity: Managed identity + least-privilege RBAC
Network: Private endpoint + controlled egress
Runtime: Foundry project, models, tools, connections
Evidence: Azure Monitor logs, metrics, traces, alerts

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

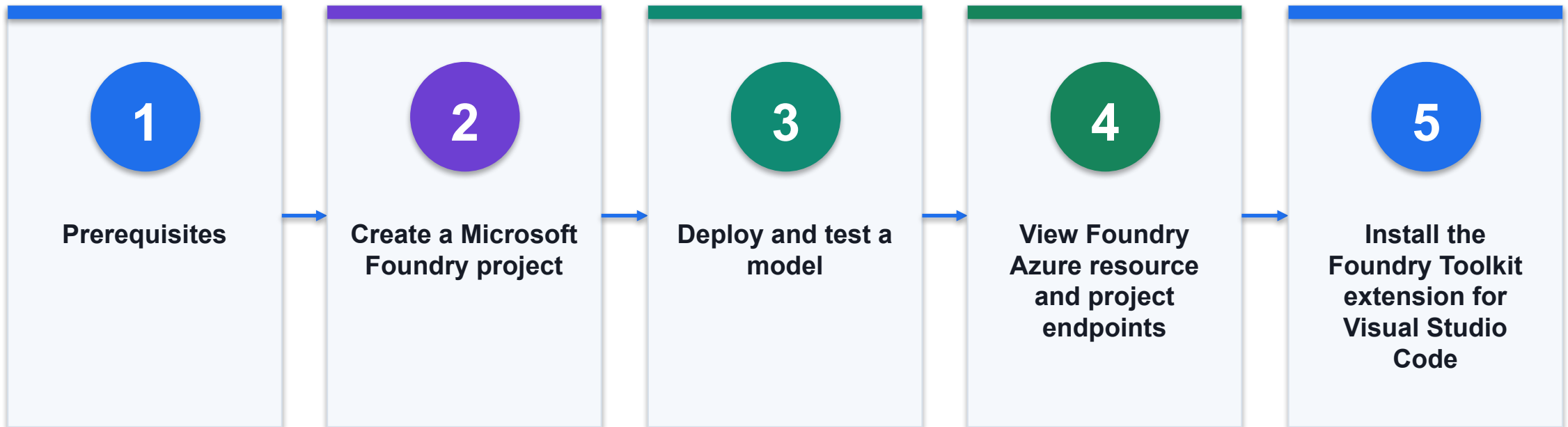
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 01 · mechanism phases



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Official lab 01 · source assets

Contract	Exact value / field	Engineering consequence
Asset 1	Instructions/Exercises/01-Explore-ai-studio.md	Exact path in pinned source
Asset 2	LICENSE	Exact path in pinned source
Asset 3	README.md	Exact path in pinned source
Asset 4	readme.md	Exact path in pinned source

Official lab 01 · evidence record

Contract	Exact value / field	Engineering consequence
Input	resource / model / source version	Capture before run
Output	result / request ID / trace	Compare to exercise expectation
Metric	quality / latency / cost	State unit and sample
Control	failure / limit / next action	Explain product implication

Lab 01: acceptance evidence

PASS WHEN

Complete the pinned Microsoft Learning exercise [official/mslearn-ai-studio/Instructions/Exercises/01-Explore-ai-studio.md](https://official.mslearn-ai-studio/Instructions/Exercises/01-Explore-ai-studio.md); capture its expected output, relevant version/endpoint, one measured result, and an explained failure or limitation.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

TOPIC 02

02

Evaluate models and release guardrails

PLAN AND MANAGE · AI-103

Evaluate models and release guardrails



Input guard

Prompt shields + content classification

Model policy

Deployment filters + approved model/version

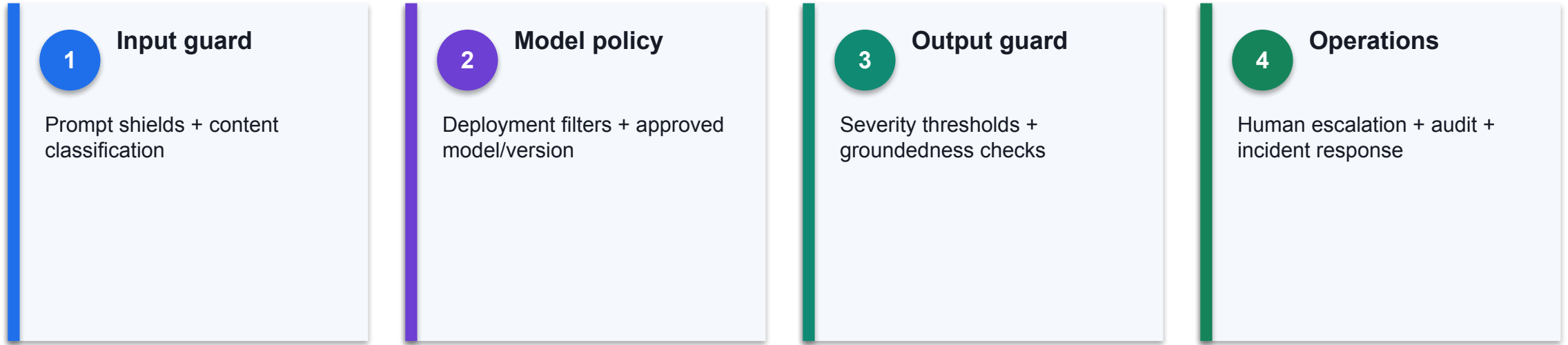
Output guard

Severity thresholds + groundedness checks

Operations

Human escalation + audit + incident response

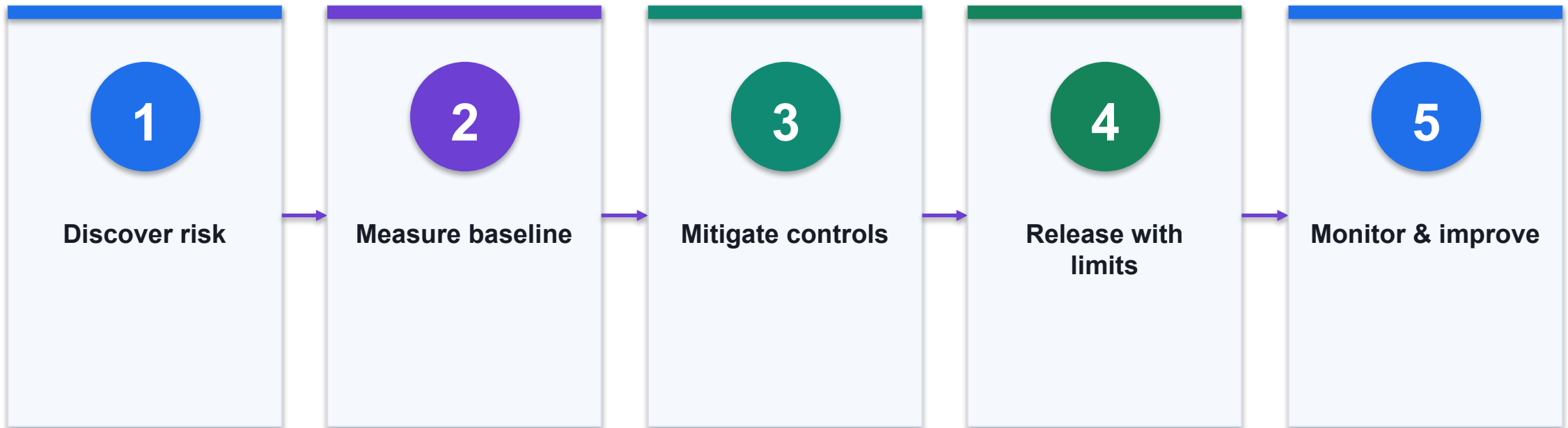
Evaluate models and release guardrails · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Evaluate models and release guardrails · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Evaluate models and release guardrails · exact contract

Contract	Exact value / field	Engineering consequence
Hate / Sexual / Violence / Self-harm	severity 0–6	Block at policy threshold
Prompt attack	direct / indirect	Reject, sanitize, or isolate tools
PII	category + offset + confidence	Redact or route to reviewer
Decision record	model, dataset, metric, owner	Versioned audit evidence

Evaluate models and release guardrails · implementation excerpt

```
POST /contentsafety/text:analyze?api-version=2024-09-01
{
  "text": user_text,
  "categories": ["Hate", "Violence", "SelfHarm",
"Sexual"],
  "outputType": "FourSeverityLevels"
}
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

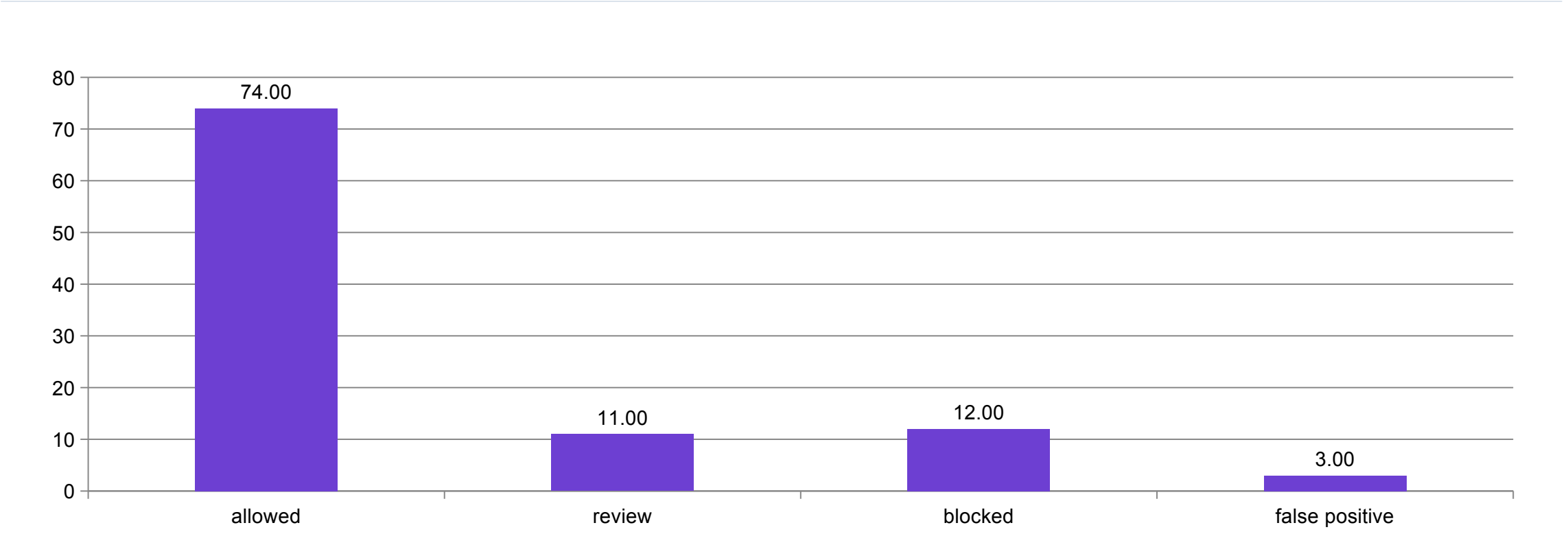
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

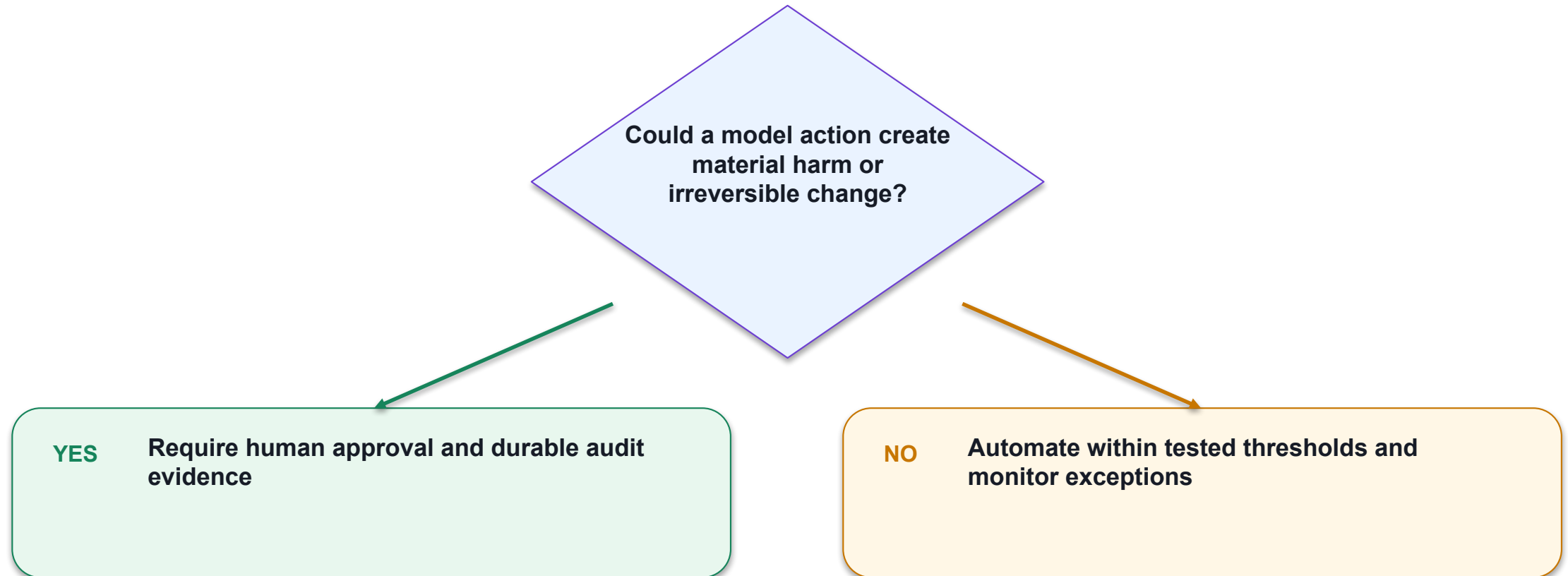
Handle auth, quota, timeout, and invalid-state paths.

Evaluate models and release guardrails · engineering metric



WHAT THE DATA SHOWS Indicative evaluation set: tune policy against both unsafe misses and legitimate-content false positives; one aggregate safety score is insufficient.

Evaluate models and release guardrails · decision rule



Evaluate models and release guardrails · failure and control

 Over-blocking LIKELY CAUSE Threshold too conservative CONTROL / RECOVERY Segment policy by channel and risk	 Jailbreak LIKELY CAUSE Instruction hierarchy compromised CONTROL / RECOVERY Isolate untrusted content and tools	 Data leak LIKELY CAUSE Prompt/output contains sensitive data CONTROL / RECOVERY Redact, minimize retention, restrict logging	 No owner LIKELY CAUSE Incident cannot be escalated CONTROL / RECOVERY Name response owner and SLA
--	--	---	--

A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 02: Evaluate models and release guardrails

Mechanism focus: Evaluate models and release guardrails



YOU WILL PRODUCE

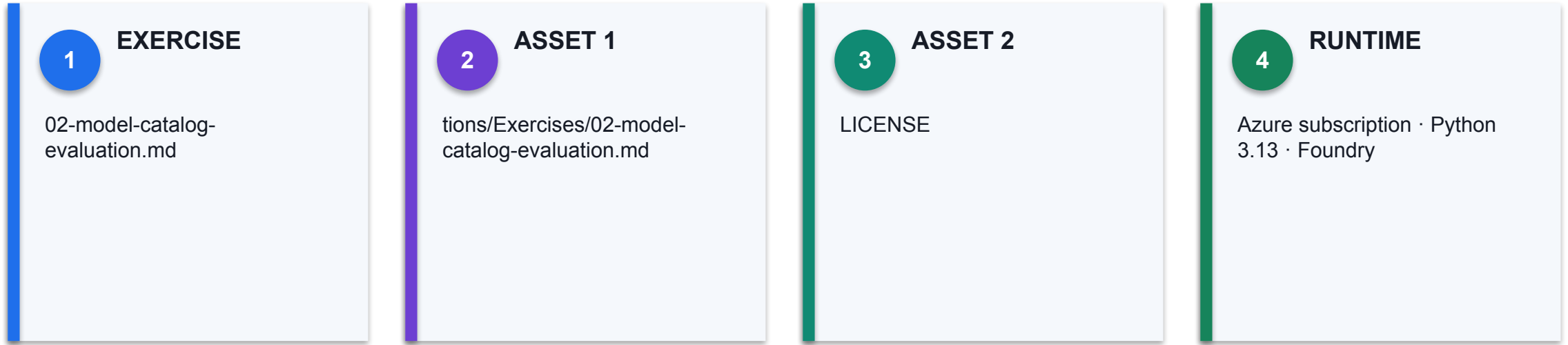
The official exercise completes with its expected output; a second learner can reproduce the result from the source instructions and the recorded environment.

Complete click-by-click procedure: Learner Guide · Lab 02 · [labs/lab-02-evaluate-models-and-release-guardrails/README.md](#)

Official lab 02 · pinned source

Contract	Exact value / field	Engineering consequence
Repository	mslearn-ai-studio	MIT source snapshot
Revision	b702dc2fd9f9	SHA-256 per file in source manifest
Exercise	02-model-catalog-evaluation.md	Unchanged instruction file
Domain	Plan and manage	AI-103 study guide

Official lab 02 · input boundaries



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Official lab 02 · implementation 1

```
POST /contentsafety/text:analyze?api-version=2024-09-01
{
  "text": user_text,
  "categories": ["Hate", "Violence", "SelfHarm",
"Sexual"],
  "outputType": "FourSeverityLevels"
}
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 02 · implementation 2

```
Hate / Sexual / Violence / Self-harm: severity 0–6 #  
Block at policy threshold  
Prompt attack: direct / indirect # Reject, sanitize,  
or isolate tools  
PII: category + offset + confidence # Redact or route  
to reviewer  
Decision record: model, dataset, metric, owner #  
Versioned audit evidence
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 02 · implementation 3

Input guard: Prompt shields + content classification
Model policy: Deployment filters + approved model/version
Output guard: Severity thresholds + groundedness checks
Operations: Human escalation + audit + incident response

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

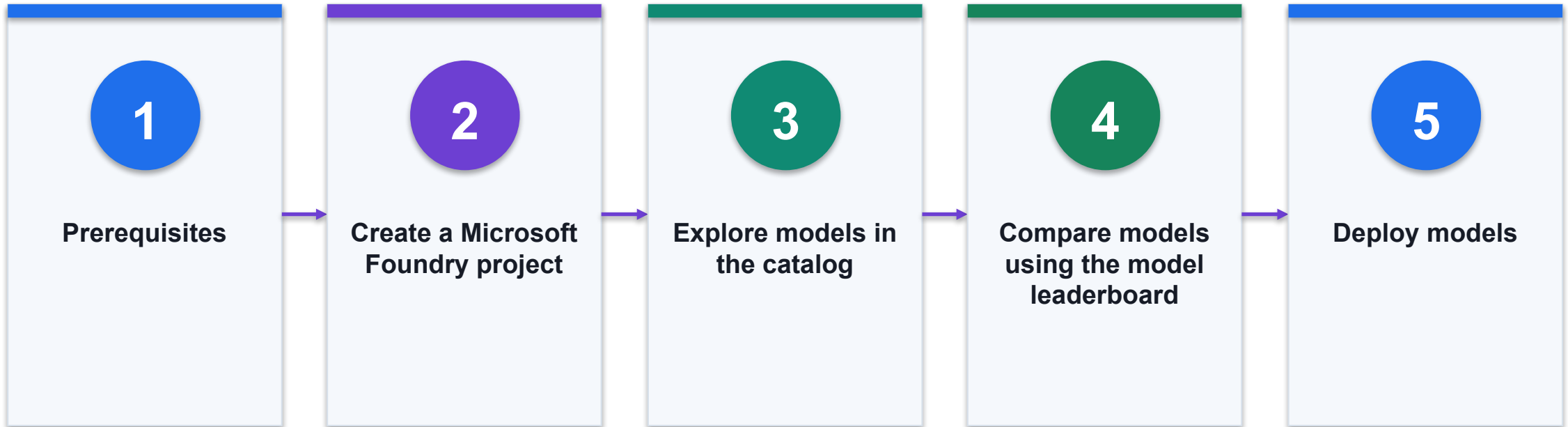
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 02 · mechanism phases



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Official lab 02 · source assets

Contract	Exact value / field	Engineering consequence
Asset 1	Instructions/Exercises/02-model-catalog-evaluation.md	Exact path in pinned source
Asset 2	LICENSE	Exact path in pinned source
Asset 3	README.md	Exact path in pinned source
Asset 4	readme.md	Exact path in pinned source

Official lab 02 · evidence record

Contract	Exact value / field	Engineering consequence
Input	resource / model / source version	Capture before run
Output	result / request ID / trace	Compare to exercise expectation
Metric	quality / latency / cost	State unit and sample
Control	failure / limit / next action	Explain product implication

Lab 02: acceptance evidence

PASS WHEN

Complete the pinned Microsoft Learning exercise [official/mslearn-ai-studio/Instructions/Exercises/02-model-catalog-evaluation.md](https://github.com/microsoft/ai-studio/blob/main/Instructions/Exercises/02-model-catalog-evaluation.md); capture its expected output, relevant version/endpoint, one measured result, and an explained failure or limitation.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

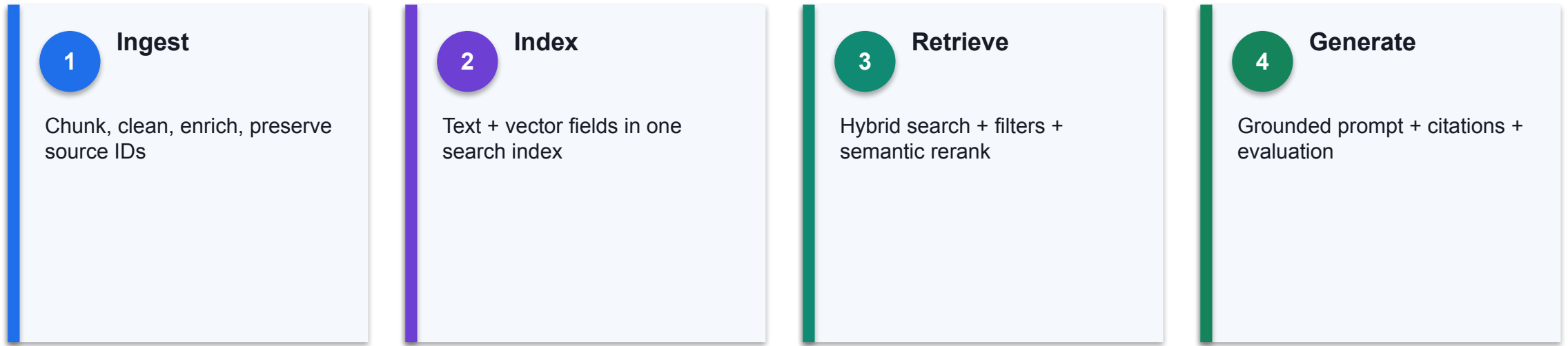
TOPIC 03

03

Build a generative chat app

GENERATIVE AI AND AGENTIC · AI-103

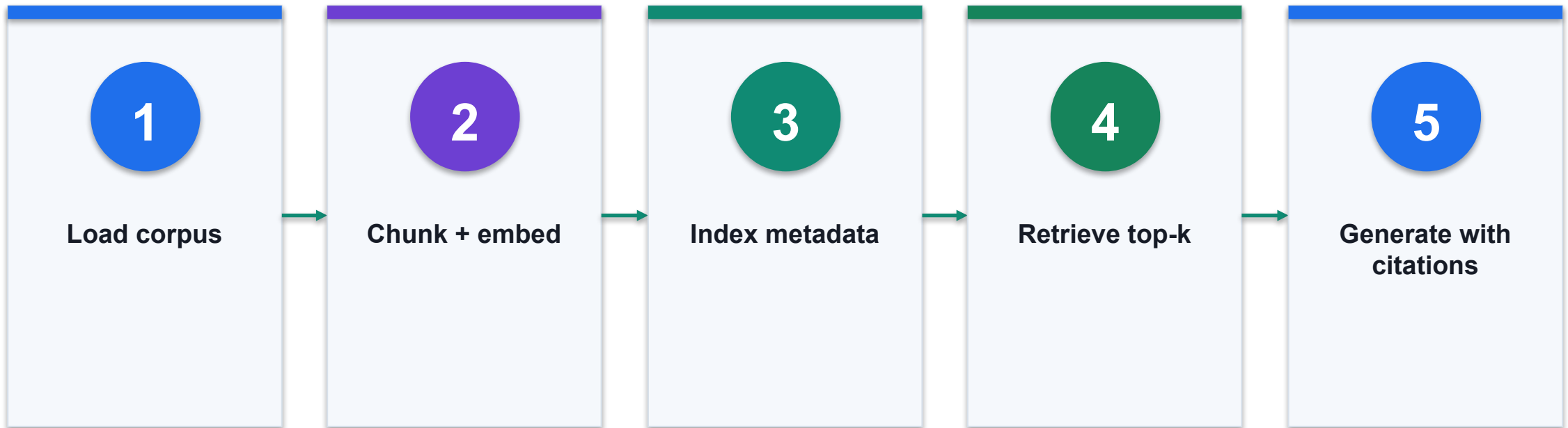
Build a generative chat app · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Build a generative chat app · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Build a generative chat app · exact contract

Contract	Exact value / field	Engineering consequence
chunk_id	string, key	Stable evidence identifier
content_vector	Collection(Edm.Single)	Dimensions match embedding model
filter	tenant_id eq 'T-42'	Security boundary before generation
top	5–20	Recall/latency trade-off

Build a generative chat app · implementation excerpt

```
response = openai.responses.create(
    model=DEPLOYMENT,
    input=[{"role": "user", "content": question}],
    tools=[{"type": "file_search",
            "vector_store_ids": [STORE_ID]}]
)
print(response.output_text)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

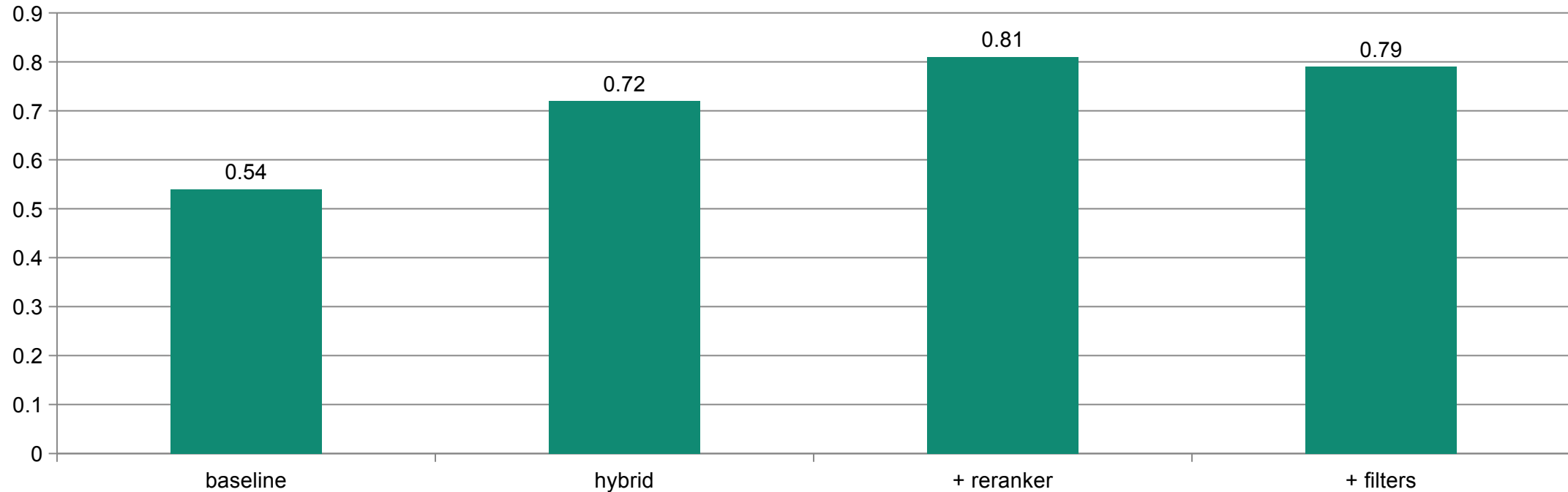
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

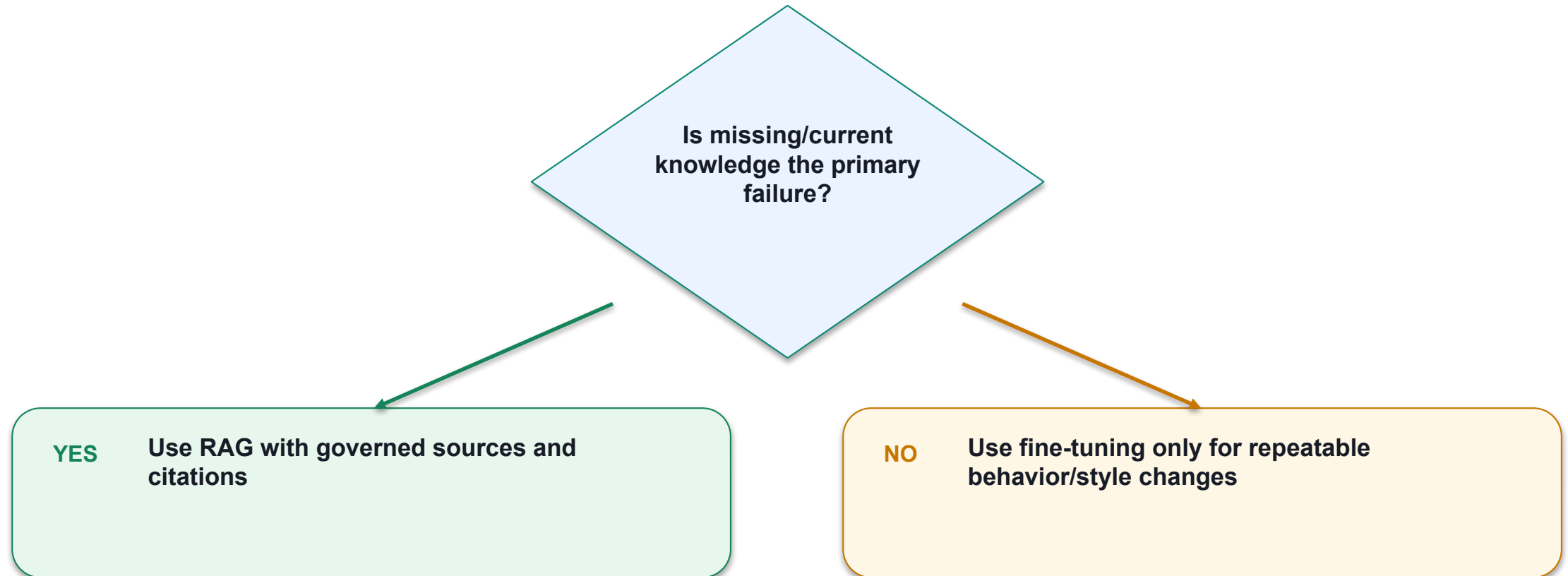
Build a generative chat app · engineering metric



WHAT THE DATA SHOWS

Indicative grounded-answer score: hybrid retrieval improves recall; reranking improves ordering; security filters may reduce recall and must be measured per tenant.

Build a generative chat app · decision rule



Build a generative chat app · failure and control

 Hallucination LIKELY CAUSE Context absent or ignored CONTROL / RECOVERY Require citations; answer 'not found'	 Dimension error LIKELY CAUSE Index and query embeddings differ CONTROL / RECOVERY Use one embedding model/config	 Leakage LIKELY CAUSE Filter applied after retrieval CONTROL / RECOVERY Filter at query/index boundary	 Stale corpus LIKELY CAUSE Indexer watermark failed CONTROL / RECOVERY Track source version and last-success time
--	---	--	---

A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 03: Build a generative chat app

Mechanism focus: Build a generative chat app



YOU WILL PRODUCE

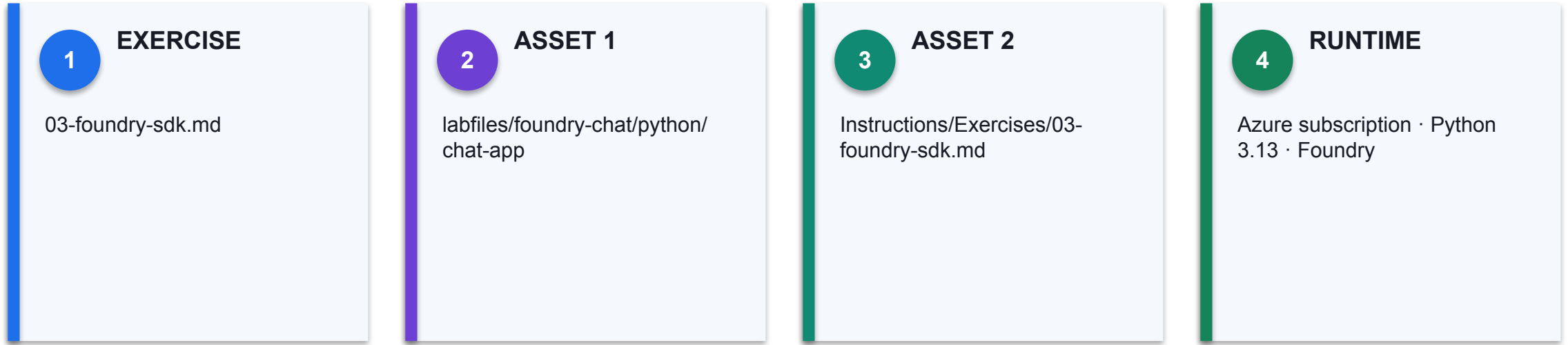
The official exercise completes with its expected output; a second learner can reproduce the result from the source instructions and the recorded environment.

Complete click-by-click procedure: Learner Guide · Lab 03 · [labs/lab-03-build-a-generative-chat-app/README.md](#)

Official lab 03 · pinned source

Contract	Exact value / field	Engineering consequence
Repository	mslearn-ai-studio	MIT source snapshot
Revision	b702dc2fd9f9	SHA-256 per file in source manifest
Exercise	03-foundry-sdk.md	Unchanged instruction file
Domain	Generative AI and agentic	AI-103 study guide

Official lab 03 · input boundaries



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Official lab 03 · implementation 1

```
# Get a response
stream = openai_client.responses.create(
    model=model_deployment,
    instructions="You are a helpful AI assistant that
answers questions and provides information.",
    input=input_text,
    previous_response_id=last_response_id,
    stream=True
)
for event in stream:
    if event.type == "response.output_text.delta":
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 03 · implementation 2

```
# Get a response
completion = openai_client.chat.completions.create(
    model=model_deployment,
    messages=[
        {
            "role": "system",
            "content": "You are a helpful AI assistant that
answers questions and provides information."
        },
        {
            "role": "user",
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 03 · implementation 3

```
# Await an asynchronous response
response = await async_client.responses.create(
    model=model_deployment,
    instructions="You are a helpful AI assistant that
answers questions and provides information.",
    input=input_text,
    previous_response_id=last_response_id
)
assistant_text = response.output_text
print("Assistant:", assistant_text)
last_response_id = response.id
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

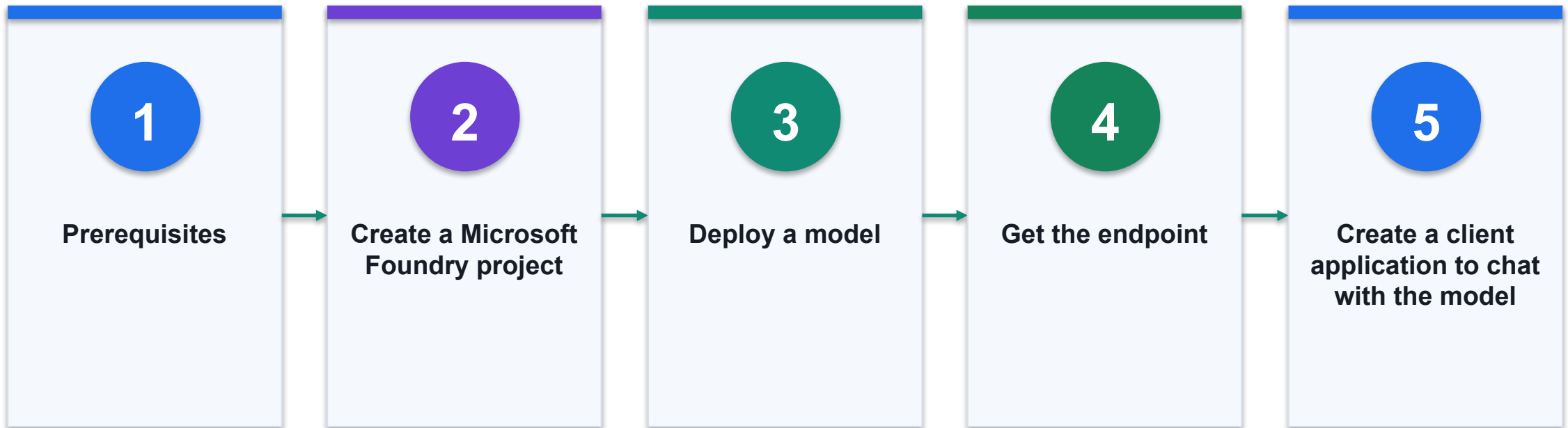
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 03 · mechanism phases



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Official lab 03 · source assets

Contract	Exact value / field	Engineering consequence
Asset 1	labfiles/foundry-chat/python/chat-app	Exact path in pinned source
Asset 2	Instructions/Exercises/03-foundry-sdk.md	Exact path in pinned source
Asset 3	LICENSE	Exact path in pinned source
Asset 4	README.md	Exact path in pinned source

Official lab 03 · evidence record

Contract	Exact value / field	Engineering consequence
Input	resource / model / source version	Capture before run
Output	result / request ID / trace	Compare to exercise expectation
Metric	quality / latency / cost	State unit and sample
Control	failure / limit / next action	Explain product implication

Lab 03: acceptance evidence

PASS WHEN

Complete the pinned Microsoft Learning exercise [official/mslearn-ai-studio/Instructions/Exercises/03-foundry-sdk.md](https://official.mslearn-ai-studio/Instructions/Exercises/03-foundry-sdk.md); capture its expected output, relevant version/endpoint, one measured result, and an explained failure or limitation.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

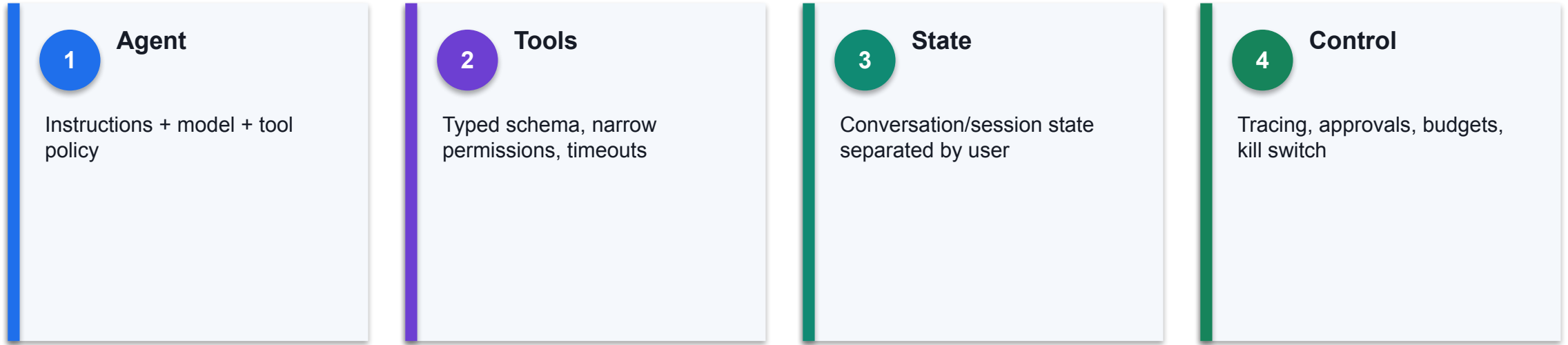
TOPIC 04

04

Build a Foundry agent

GENERATIVE AI AND AGENTIC · AI-103

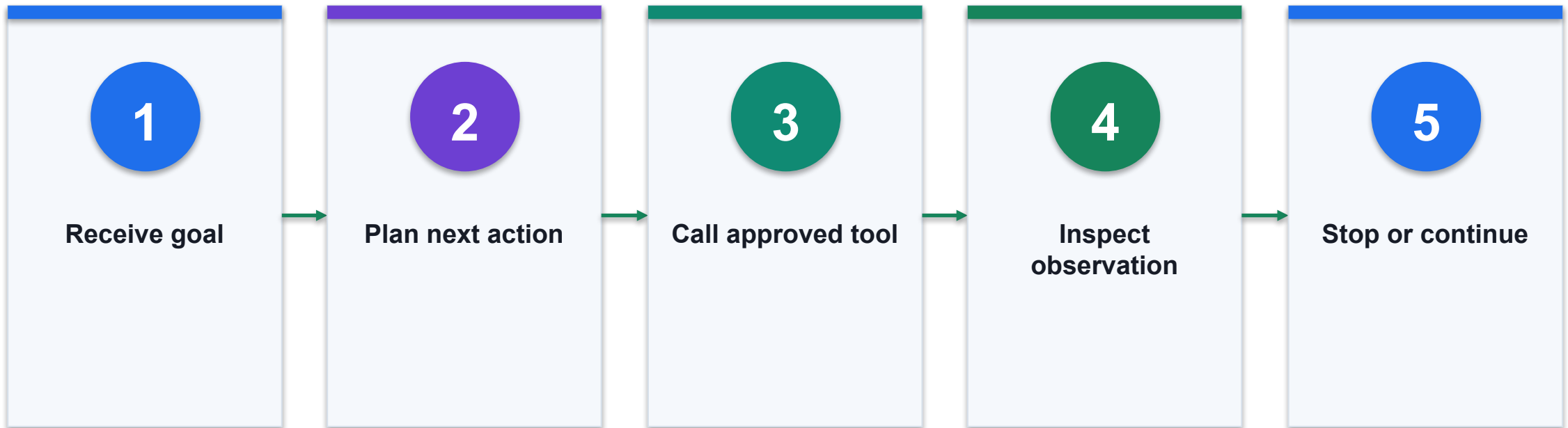
Build a Foundry agent · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Build a Foundry agent · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Build a Foundry agent · exact contract

Contract	Exact value / field	Engineering consequence
Tool schema	name + JSON parameters	Reject unknown arguments
Identity	Dedicated agent principal	No shared admin identity
Budget	max turns / tokens / cost	Hard termination condition
Trace	span, tool call, result, decision	Replayable evidence

Build a Foundry agent · implementation excerpt

```
agent = project.agents.create_version(
    agent_name="support-triage",
    definition=PromptAgentDefinition(
        model=MODEL_DEPLOYMENT,
        instructions=SYSTEM_POLICY,
        tools=[search_tool, ticket_tool]))
print(agent.id, agent.version)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

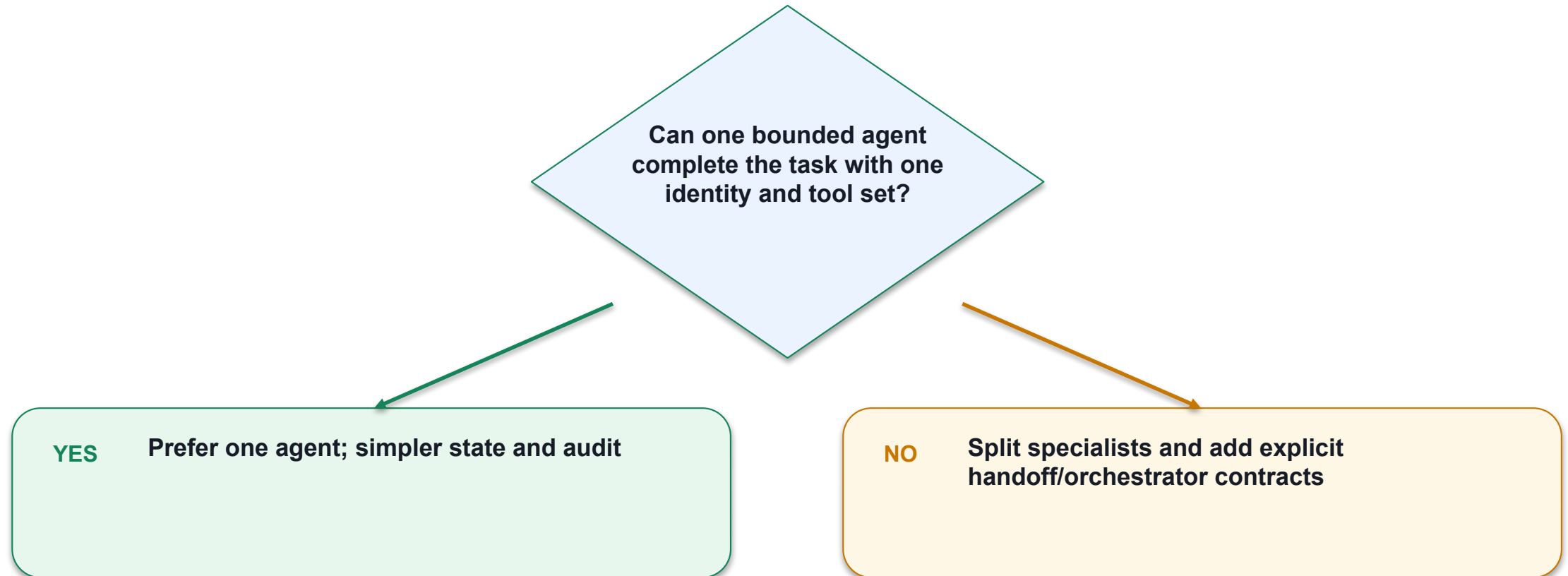
Handle auth, quota, timeout, and invalid-state paths.

Build a Foundry agent · engineering metric



WHAT THE DATA SHOWS Indicative operational complexity index: add hosting and orchestration only when autonomy, custom code, or specialization justifies extra identity, state, and observability controls.

Build a Foundry agent · decision rule



Build a Foundry agent · failure and control

 Tool abuse LIKELY CAUSE Broad permissions or weak schema CONTROL / RECOVERY Allow-list actions and validate arguments	 Loop LIKELY CAUSE No stop condition CONTROL / RECOVERY Turn/time/cost budgets	 Cross-user state LIKELY CAUSE Memory partition failure CONTROL / RECOVERY Key state by tenant + user + session	 Opaque handoff LIKELY CAUSE Agent loses context/owner CONTROL / RECOVERY Typed handoff payload + trace link
--	--	---	--

A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 04: Build a Foundry agent

Mechanism focus: Build a Foundry agent



YOU WILL PRODUCE

The official exercise completes with its expected output; a second learner can reproduce the result from the source instructions and the recorded environment.

Complete click-by-click procedure: [Learner Guide · Lab 04 · labs/lab-04-build-a-foundry-agent/README.md](#)

Official lab 04 · pinned source

Contract	Exact value / field	Engineering consequence
Repository	mslearn-ai-agents	MIT source snapshot
Revision	7eafc3c339d3	SHA-256 per file in source manifest
Exercise	01-build-agent-portal-and-vscode.md	Unchanged instruction file
Domain	Generative AI and agentic	AI-103 study guide

Official lab 04 · input boundaries

1

EXERCISE

01-build-agent-portal-and-vscode.md

2

ASSET 1

01-build-agent-portal-and-vscode/IT_Policy.txt

3

ASSET 2

agent-portal-and-vscode/system_performance.csv

4

RUNTIME

Azure subscription · Python 3.13 · Foundry

ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Official lab 04 · implementation 1

```
import base64
import os
from pathlib import Path

from azure.ai.projects import AIProjectClient
from azure.identity import DefaultAzureCredential
from dotenv import load_dotenv

OUTPUT_DIR = Path("agent_outputs")
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 04 · implementation 2

```
python -m venv labenv
.\labenv\Scripts\Activate.ps1
pip install -r requirements.txt
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 04 · implementation 3

```
agent = project.agents.create_version(  
    agent_name="support-triage",  
    definition=PromptAgentDefinition(  
        model=MODEL_DEPLOYMENT,  
        instructions=SYSTEM_POLICY,  
        tools=[search_tool, ticket_tool]))  
print(agent.id, agent.version)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

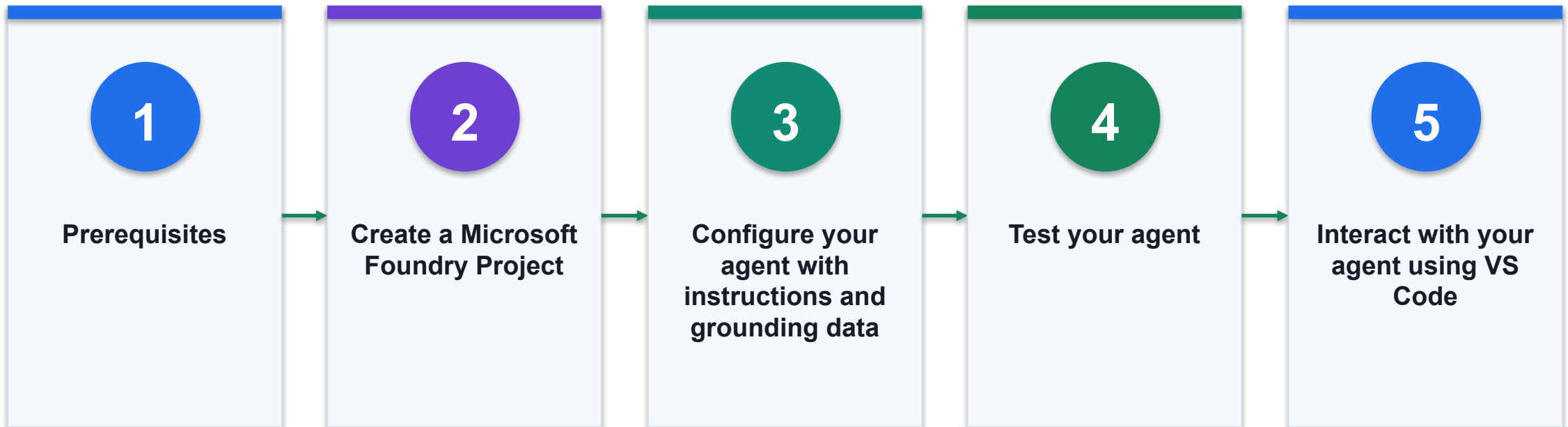
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 04 · mechanism phases



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Official lab 04 · source assets

Contract	Exact value / field	Engineering consequence
Asset 1	Labfiles/01-build-agent-portal-and-vscode/IT_Policy.txt	Exact path in pinned source
Asset 2	Labfiles/01-build-agent-portal-and-vscode/ system_performance.csv	Exact path in pinned source
Asset 3	Labfiles/01-build-agent-portal-and-vscode/Python	Exact path in pinned source
Asset 4	Instructions/Exercises/01-build-agent-portal-and- vscode.md	Exact path in pinned source

Official lab 04 · evidence record

Contract	Exact value / field	Engineering consequence
Input	resource / model / source version	Capture before run
Output	result / request ID / trace	Compare to exercise expectation
Metric	quality / latency / cost	State unit and sample
Control	failure / limit / next action	Explain product implication

Lab 04: acceptance evidence

PASS WHEN

Complete the pinned Microsoft Learning exercise [official/mslearn-ai-agents/Instructions/Exercises/01-build-agent-portal-and-vscode.md](#); capture its expected output, relevant version/endpoint, one measured result, and an explained failure or limitation.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

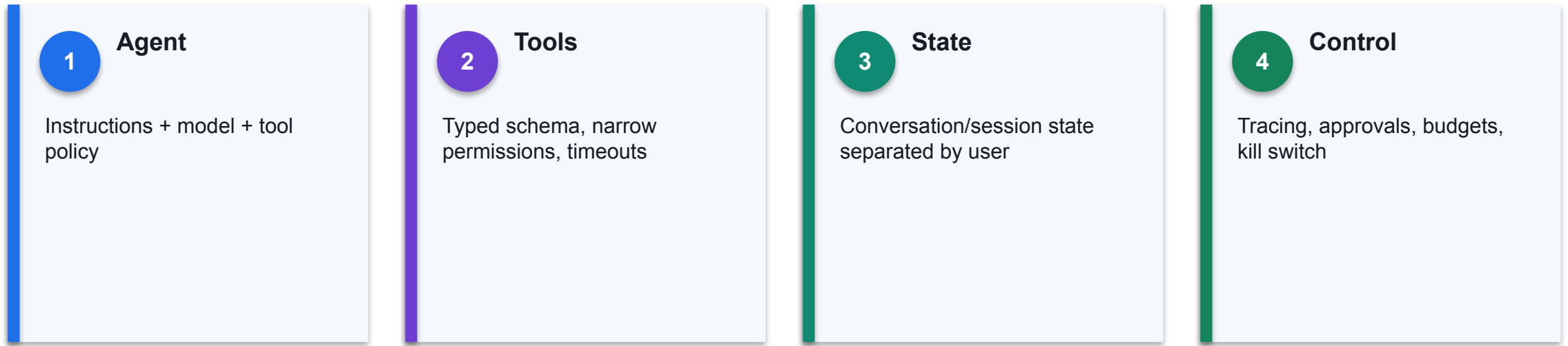
TOPIC 05

05

Use a custom tool in an agent

GENERATIVE AI AND AGENTIC · AI-103

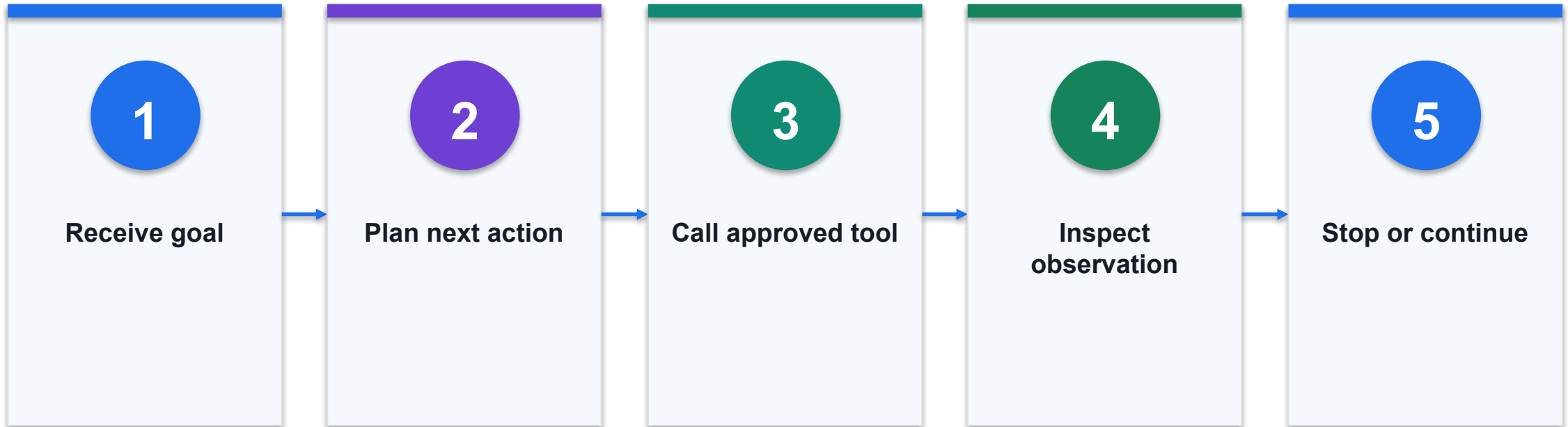
Use a custom tool in an agent - architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Use a custom tool in an agent - runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Use a custom tool in an agent - exact contract

Contract	Exact value / field	Engineering consequence
Tool schema	name + JSON parameters	Reject unknown arguments
Identity	Dedicated agent principal	No shared admin identity
Budget	max turns / tokens / cost	Hard termination condition
Trace	span, tool call, result, decision	Replayable evidence

Use a custom tool in an agent · implementation excerpt

```
agent = project.agents.create_version(
    agent_name="support-triage",
    definition=PromptAgentDefinition(
        model=MODEL_DEPLOYMENT,
        instructions=SYSTEM_POLICY,
        tools=[search_tool, ticket_tool]))
print(agent.id, agent.version)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

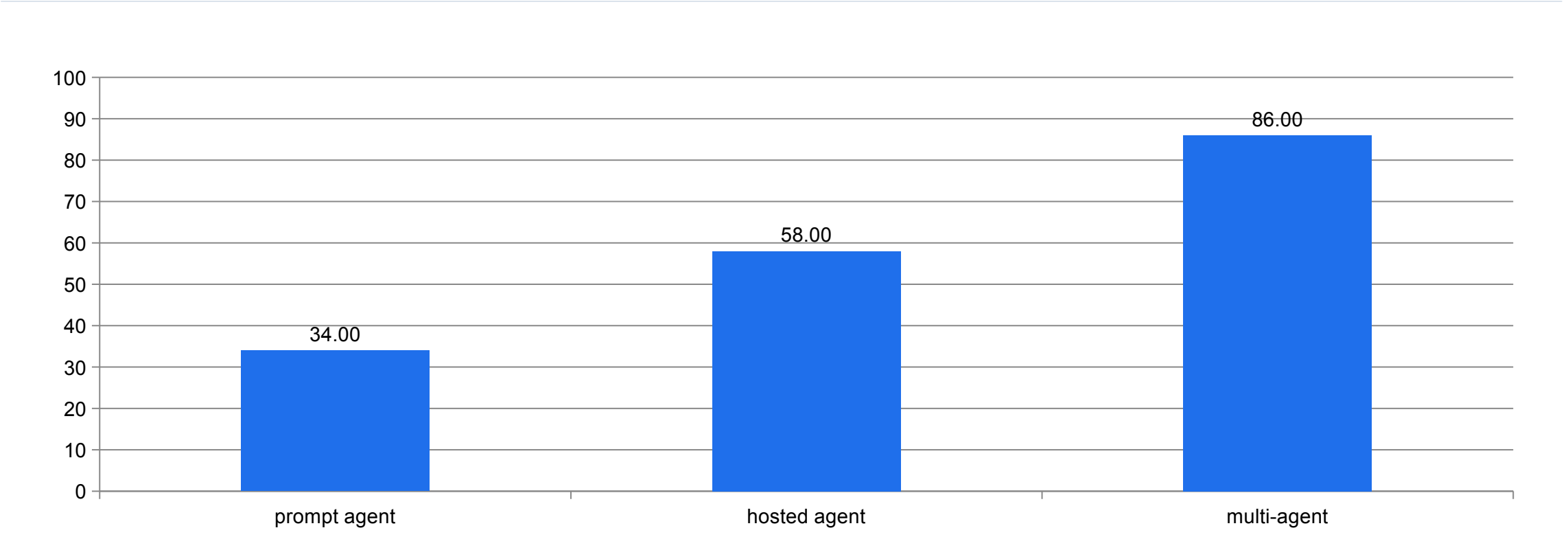
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

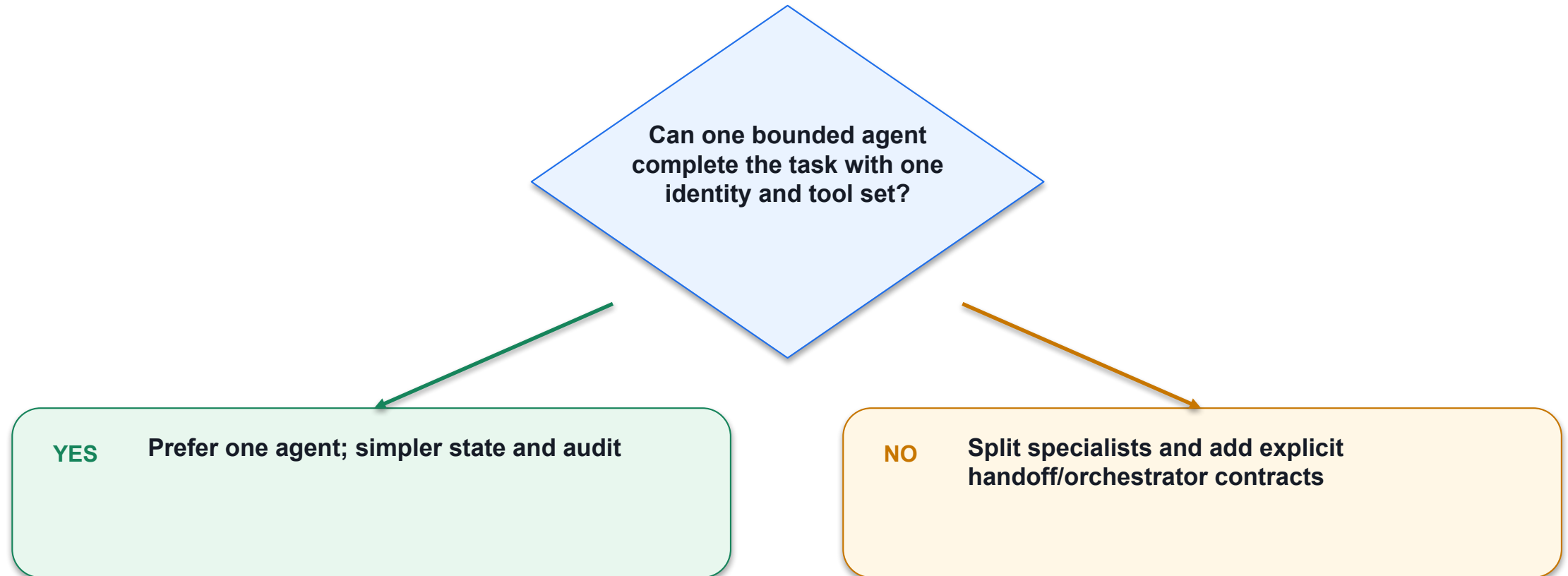
Handle auth, quota, timeout, and invalid-state paths.

Use a custom tool in an agent - engineering metric



WHAT THE DATA SHOWS Indicative operational complexity index: add hosting and orchestration only when autonomy, custom code, or specialization justifies extra identity, state, and observability controls.

Use a custom tool in an agent - decision rule



Use a custom tool in an agent - failure and control

 Tool abuse LIKELY CAUSE Broad permissions or weak schema CONTROL / RECOVERY Allow-list actions and validate arguments	 Loop LIKELY CAUSE No stop condition CONTROL / RECOVERY Turn/time/cost budgets	 Cross-user state LIKELY CAUSE Memory partition failure CONTROL / RECOVERY Key state by tenant + user + session	 Opaque handoff LIKELY CAUSE Agent loses context/owner CONTROL / RECOVERY Typed handoff payload + trace link
--	--	---	--

A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 05: Use a custom tool in an agent

Mechanism focus: Use a custom tool in an agent



YOU WILL PRODUCE

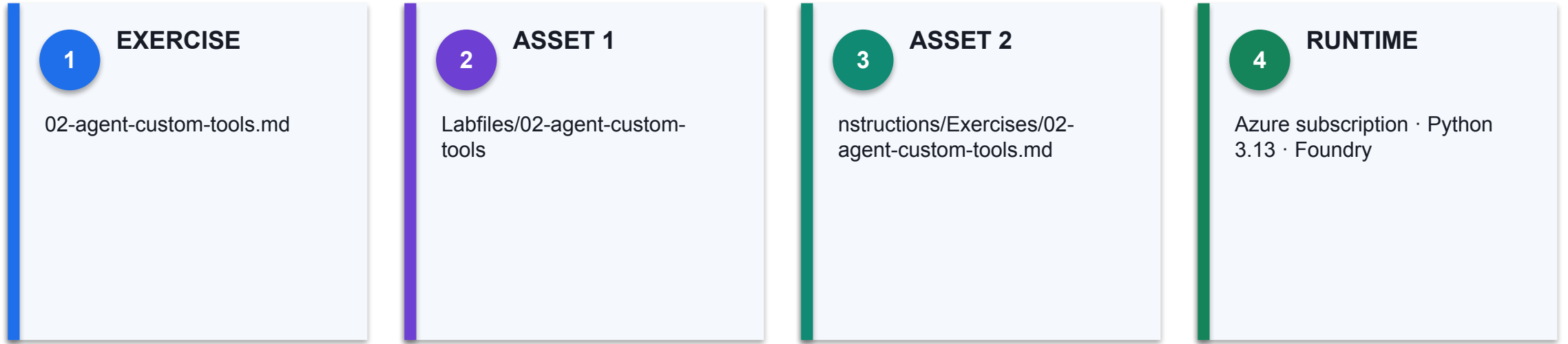
The official exercise completes with its expected output; a second learner can reproduce the result from the source instructions and the recorded environment.

Complete click-by-click procedure: [Learner Guide · Lab 05 · labs/lab-05-use-a-custom-tool-in-an-agent/README.md](#)

Official lab 05 · pinned source

Contract	Exact value / field	Engineering consequence
Repository	mslearn-ai-agents	MIT source snapshot
Revision	7eafc3c339d3	SHA-256 per file in source manifest
Exercise	02-agent-custom-tools.md	Unchanged instruction file
Domain	Generative AI and agentic	AI-103 study guide

Official lab 05 · input boundaries



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Official lab 05 · implementation 1

```
# Define the observation report generation function tool
report_tool = FunctionTool(
    name="generate_observation_report",
    description="Generate a report summarizing an astronomical
observation",
    parameters={
        "type": "object",
        "properties": {
            "event_name": {
                "type": "string",
                "description": "the name of the astronomical
event being observed",
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 05 · implementation 2

```
# Define the observation cost function tool
cost_tool = FunctionTool(
    name="calculate_observation_cost",
    description="Calculate the cost of an observation based on
the telescope tier, number of hours, and priority level.",
    parameters={
        "type": "object",
        "properties": {
            "telescope_tier": {
                "type": "string",
                "description": "the tier of the telescope (e.g.
'standard', 'advanced', 'premium')",
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 05 · implementation 3

```
# Process function calls
for item in response.output:
    if item.type == "function_call":
        # Retrieve the matching function tool
        function_name = item.name
        result = None
        if item.name == "next_visible_event":
            result =
next_visible_event(**json.loads(item.arguments))
        elif item.name == "calculate_observation_cost":
            result =
calculate_observation_cost(**json.loads(item.arguments))
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

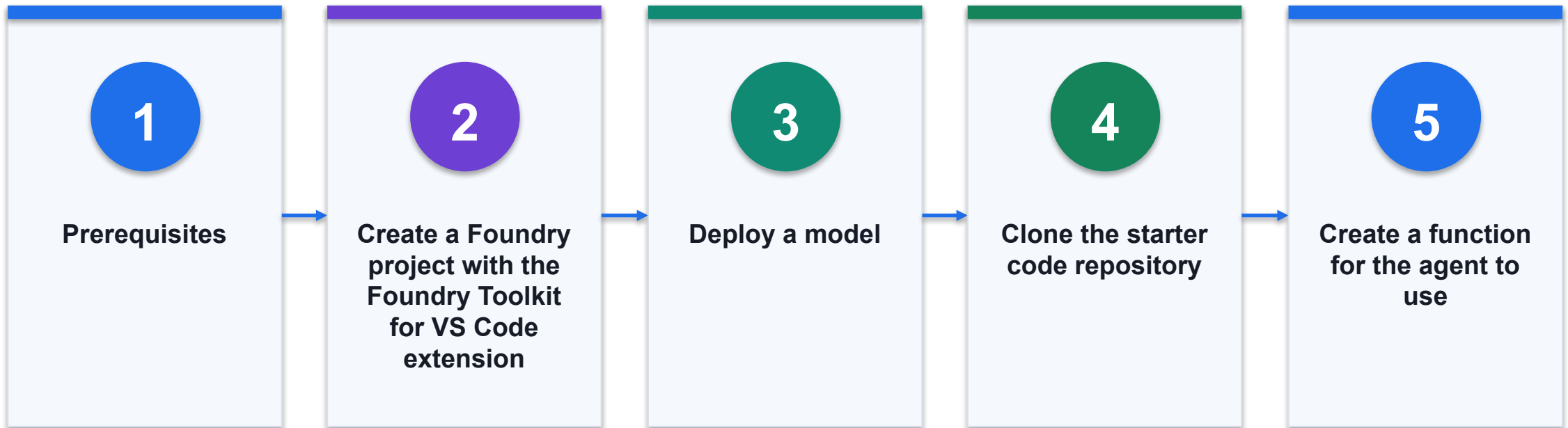
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 05 · mechanism phases



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Official lab 05 · source assets

Contract	Exact value / field	Engineering consequence
Asset 1	Labfiles/02-agent-custom-tools	Exact path in pinned source
Asset 2	Instructions/Exercises/02-agent-custom-tools.md	Exact path in pinned source
Asset 3	LICENSE	Exact path in pinned source
Asset 4	README.md	Exact path in pinned source

Official lab 05 · evidence record

Contract	Exact value / field	Engineering consequence
Input	resource / model / source version	Capture before run
Output	result / request ID / trace	Compare to exercise expectation
Metric	quality / latency / cost	State unit and sample
Control	failure / limit / next action	Explain product implication

Lab 05: acceptance evidence

PASS WHEN

Complete the pinned Microsoft Learning exercise [official/mslearn-ai-agents/Instructions/Exercises/02-agent-custom-tools.md](#); capture its expected output, relevant version/endpoint, one measured result, and an explained failure or limitation.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

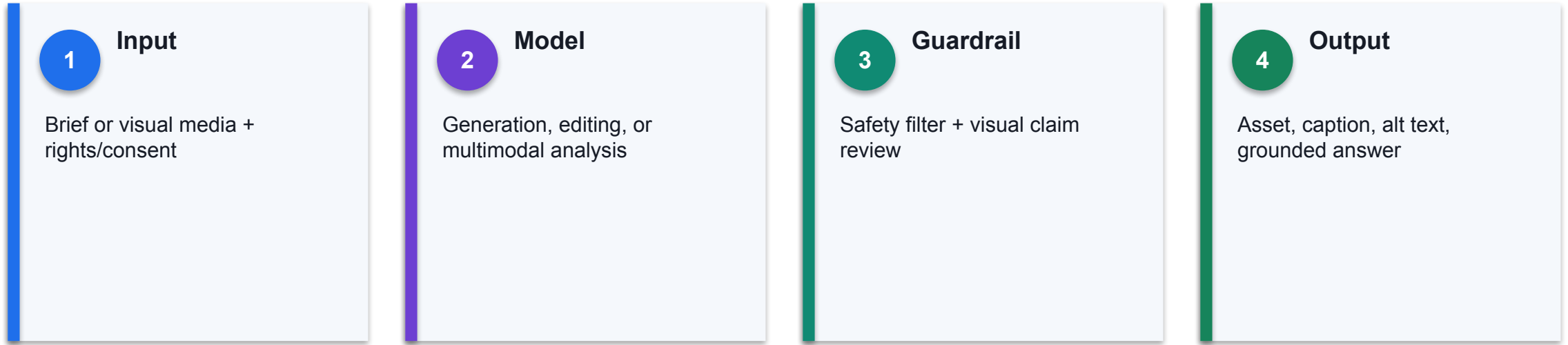
TOPIC 06

06

Develop a vision-enabled chat app

COMPUTER VISION · AI-103

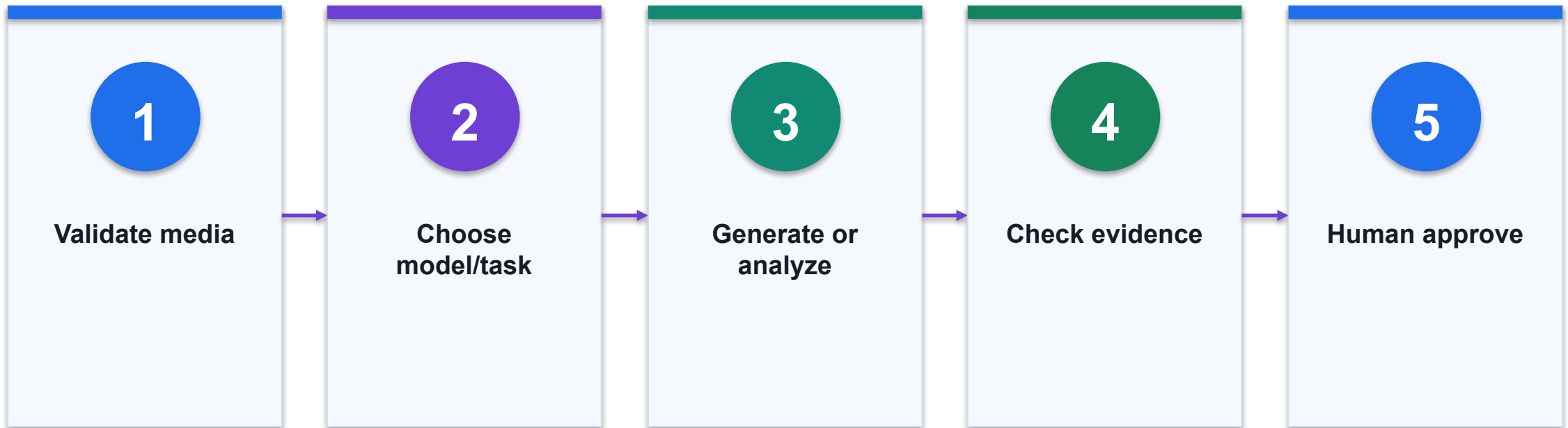
Develop a vision-enabled chat app · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Develop a vision-enabled chat app · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Develop a vision-enabled chat app · exact contract

Contract	Exact value / field	Engineering consequence
asset_id	stable source/output ID	Provenance and rollback
model_version	pinned deployment/version	Reproducible evaluation
claim	text + source evidence	Reject unsupported product facts
review	approved + reason + owner	Publication gate

Develop a vision-enabled chat app · implementation excerpt

```
brief = {"capacity_ml": 500, "allowed_claims":
["reusable"]}
claims = ["reusable", "48-hour cooling"]
unsupported = set(claims) -
set(brief["allowed_claims"])
assert not unsupported, f"Reject unsupported claims:
{unsupported}"
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

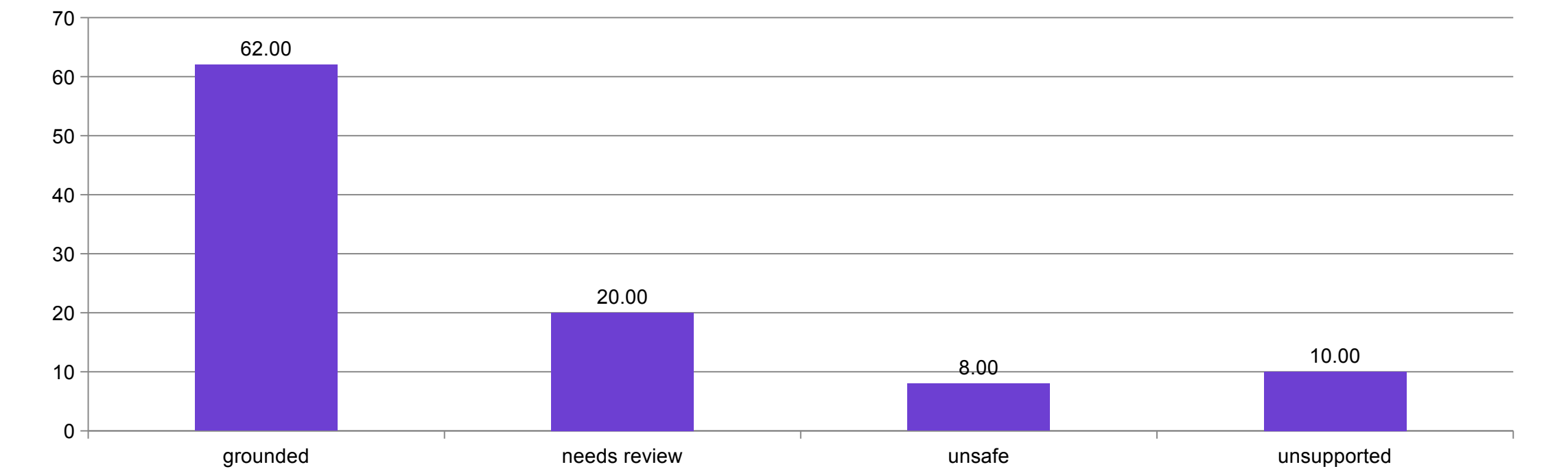
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

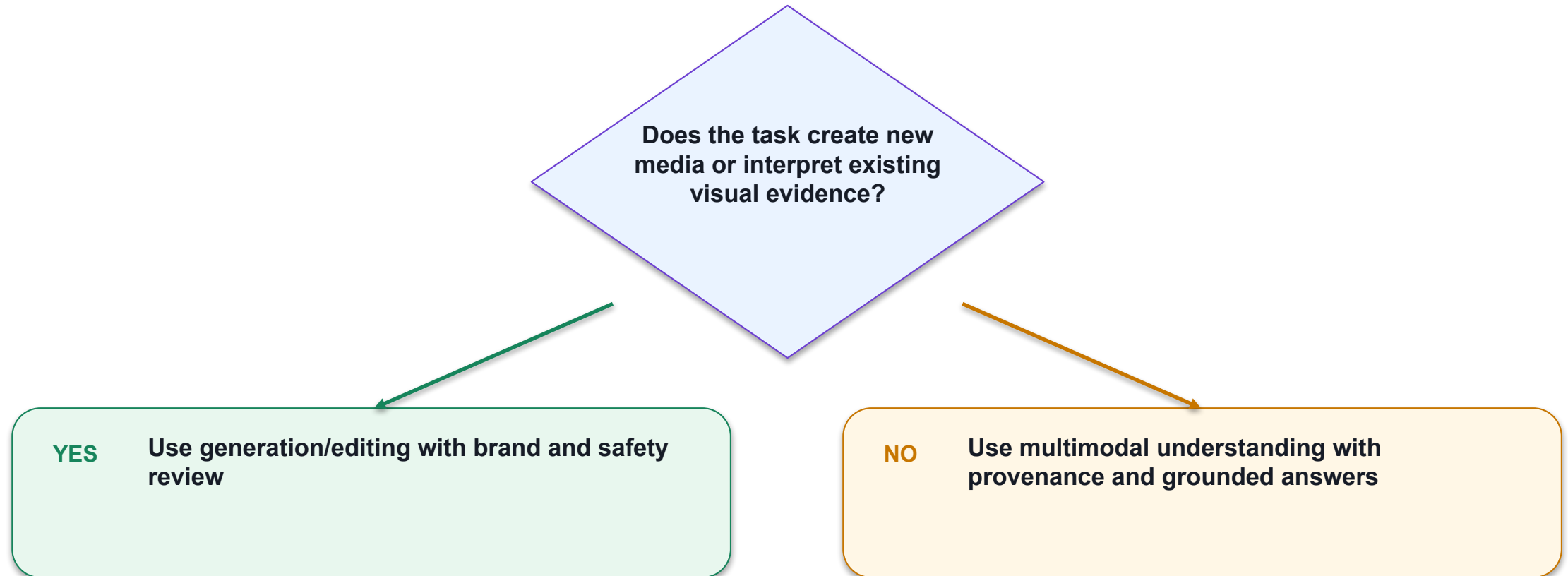
Handle auth, quota, timeout, and invalid-state paths.

Develop a vision-enabled chat app · engineering metric



WHAT THE DATA SHOWS Illustrative review set (%): generation quality alone does not grant publication; groundedness, safety, accessibility, and human approval are separate gates.

Develop a vision-enabled chat app · decision rule



Develop a vision-enabled chat app · failure and control

 Fabricated claim LIKELY CAUSE Output exceeds source brief CONTROL / RECOVERY Compare with approved attributes	 Prompt injection LIKELY CAUSE Text embedded in image instructs agent CONTROL / RECOVERY Treat media as untrusted data	 Accessibility LIKELY CAUSE Alt text omits visual facts CONTROL / RECOVERY Evaluate against source image	 Rights LIKELY CAUSE Reference media lacks permission CONTROL / RECOVERY Validate provenance and consent
--	--	--	--

A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 06: Develop a vision-enabled chat app

Mechanism focus: Develop a vision-enabled chat app



YOU WILL PRODUCE

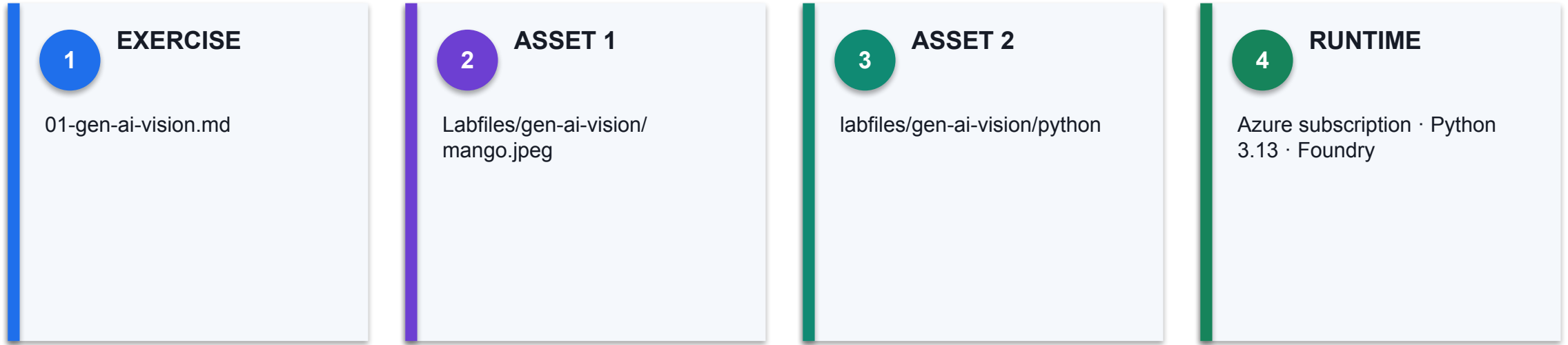
The official exercise completes with its expected output; a second learner can reproduce the result from the source instructions and the recorded environment.

Complete click-by-click procedure: [Learner Guide · Lab 06 · labs/lab-06-develop-a-vision-enabled-chat-app/README.md](#)

Official lab 06 · pinned source

Contract	Exact value / field	Engineering consequence
Repository	mslearn-ai-vision	MIT source snapshot
Revision	5d8e9d8a2ce4	SHA-256 per file in source manifest
Exercise	01-gen-ai-vision.md	Unchanged instruction file
Domain	Computer vision	AI-103 study guide

Official lab 06 · input boundaries



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Official lab 06 · implementation 1

```
# Get a response to image input
image_path = Path("mystery-fruit.jpeg")
image_format = "jpeg"
with open(image_path, "rb") as image_file:
    image_data =
base64.b64encode(image_file.read()).decode("utf-8")

data_url = f"data:image/{image_format};base64,{image_data}"

response = client.responses.create(
    model=model_deployment,
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 06 · implementation 2

```
# Get a response to image input
image_url = "https://microsoftlearning.github.io/mslearn-ai-vision/Labfiles/gen-ai-vision/orange.jpeg"
response = client.responses.create(
    model=model_deployment,
    input=[
        {"role": "developer", "content": system_message},
        { "role": "user", "content": [
            { "type": "input_text", "text": prompt},
            { "type": "input_image", "image_url": image_url}
        ]}
    ]
)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 06 · implementation 3

```
# Create an OpenAI client
credential = DefaultAzureCredential()
token_provider =
get_bearer_token_provider(credential,
"https://ai.azure.com/.default")
client = OpenAI(
    base_url=openai_endpoint,
    api_key=token_provider()
)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

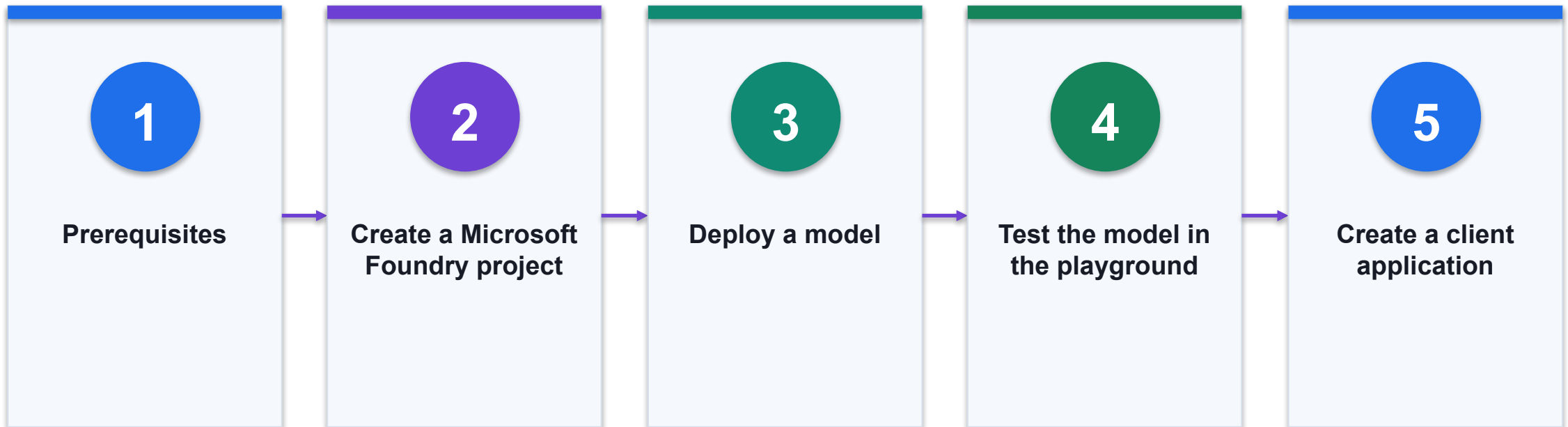
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 06 · mechanism phases



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Official lab 06 · source assets

Contract	Exact value / field	Engineering consequence
Asset 1	Labfiles/gen-ai-vision/mango.jpeg	Exact path in pinned source
Asset 2	labfiles/gen-ai-vision/python	Exact path in pinned source
Asset 3	Labfiles/gen-ai-vision/python	Exact path in pinned source
Asset 4	Labfiles/gen-ai-vision/orange.jpeg	Exact path in pinned source

Official lab 06 · evidence record

Contract	Exact value / field	Engineering consequence
Input	resource / model / source version	Capture before run
Output	result / request ID / trace	Compare to exercise expectation
Metric	quality / latency / cost	State unit and sample
Control	failure / limit / next action	Explain product implication

Lab 06: acceptance evidence

PASS WHEN

Complete the pinned Microsoft Learning exercise [official/mslearn-ai-vision/Instructions/Exercises/01-gen-ai-vision.md](#); capture its expected output, relevant version/endpoint, one measured result, and an explained failure or limitation.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

TOPIC 00

00

Day 2

Text analysis · agents · content understanding · knowledge mining

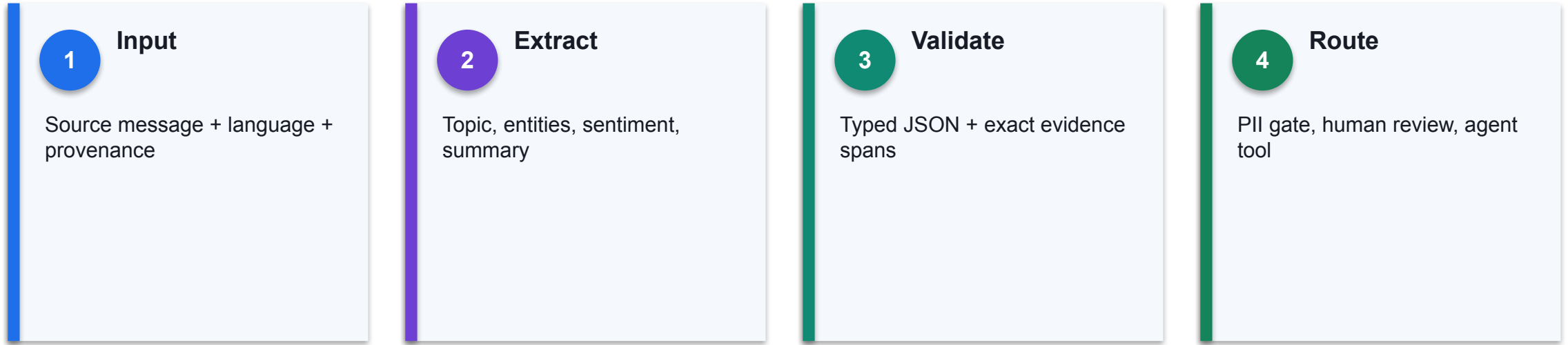
TOPIC 07

07

Analyze text

TEXT ANALYSIS · AI-103

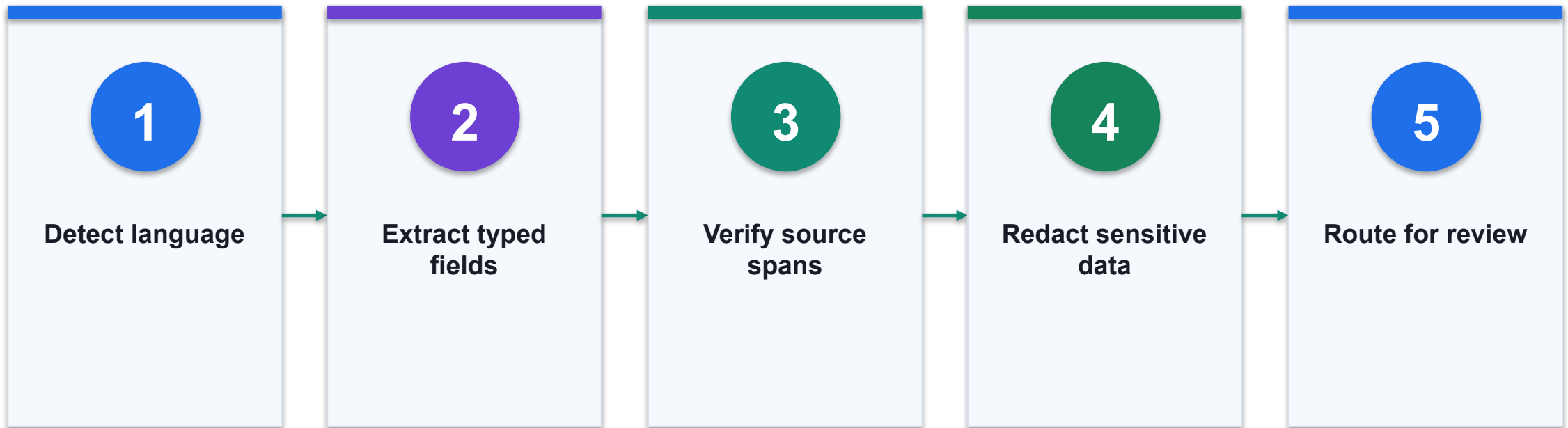
Analyze text · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Analyze text · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Analyze text · exact contract

Contract	Exact value / field	Engineering consequence
message_id	stable source ID	Traceability
evidence_span	exact source substring	Reject unsupported claims
sensitive_content	boolean	Review and redact
review_required	boolean	No unsafe auto-action

Analyze text · implementation excerpt

```
record = {"message_id": "M1", "evidence_span": "refund  
for order 12345"}  
source = "I need a refund for order 12345."  
assert record["evidence_span"] in source  
# Validate every generated field before an agent uses  
it
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

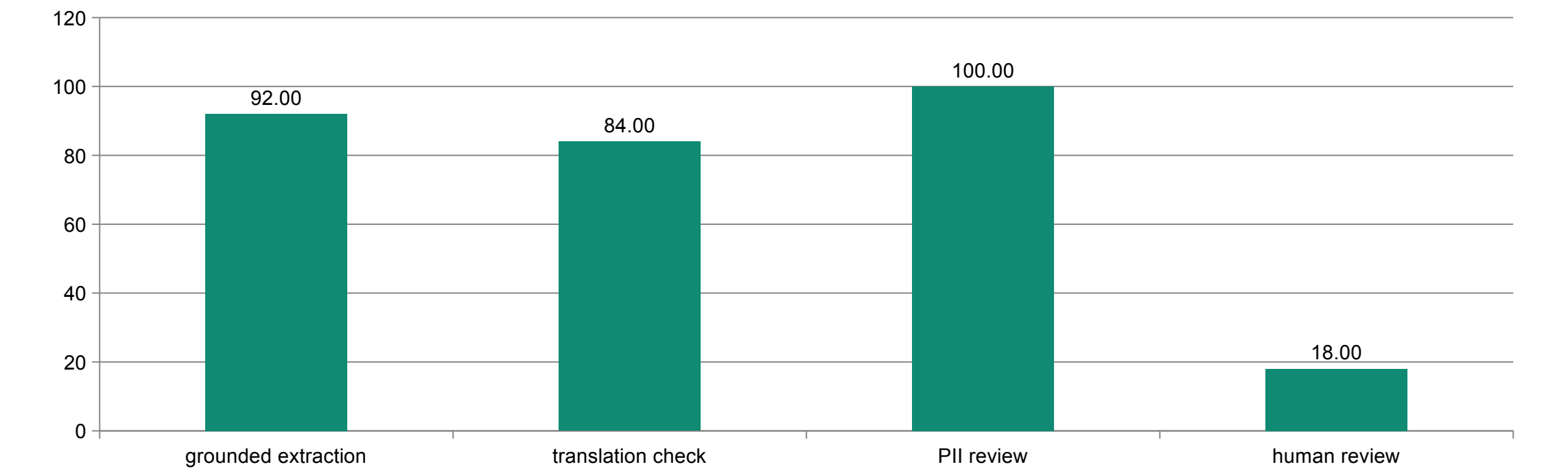
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

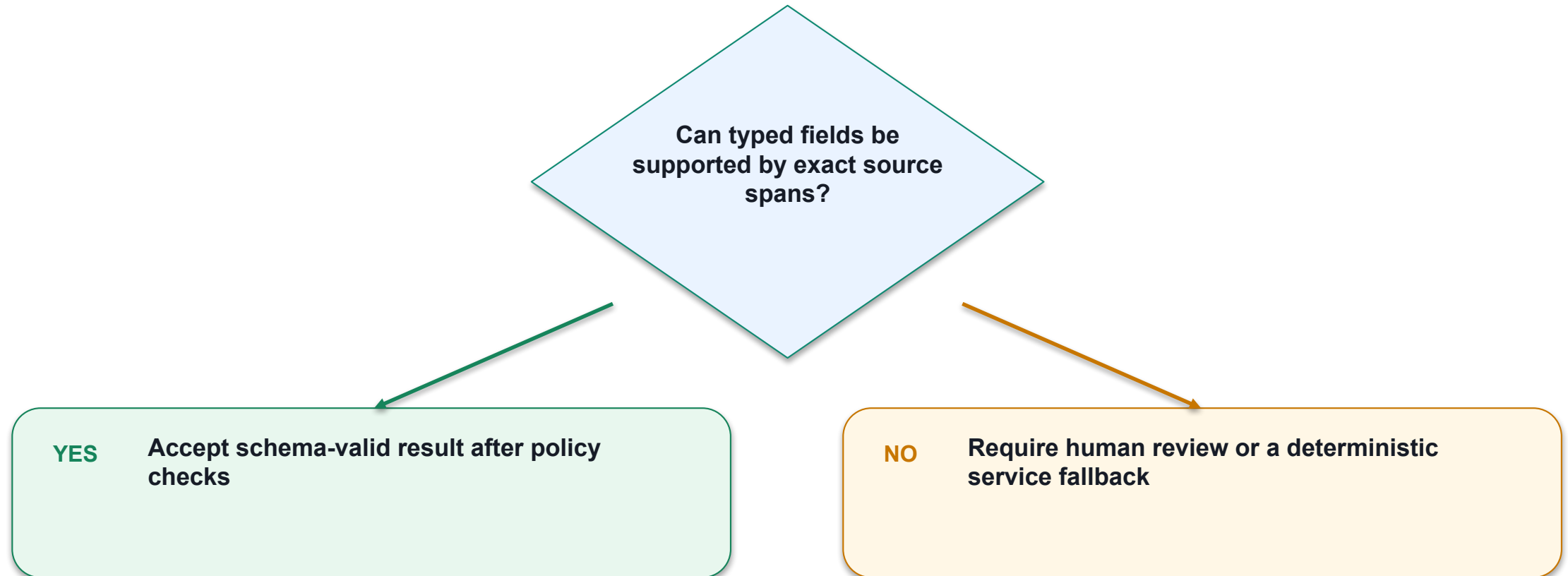
Handle auth, quota, timeout, and invalid-state paths.

Analyze text · engineering metric



WHAT THE DATA SHOWS Illustrative evaluation set (%): report source-grounded extraction, translation fidelity, and sensitive-content handling separately.

Analyze text · decision rule



Analyze text · failure and control



Invented entity

LIKELY CAUSE

Model guesses a customer fact

CONTROL / RECOVERY

Require exact evidence span



PII leak

LIKELY CAUSE

Identifier enters logs or tool

CONTROL / RECOVERY

Redact before routing



Translation drift

LIKELY CAUSE

Domain term changes meaning

CONTROL / RECOVERY

Compare source and translation



Bad JSON

LIKELY CAUSE

Agent tool receives malformed output

CONTROL / RECOVERY

Validate types and required fields

A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 07: Analyze text

Mechanism focus: Analyze text



YOU WILL PRODUCE

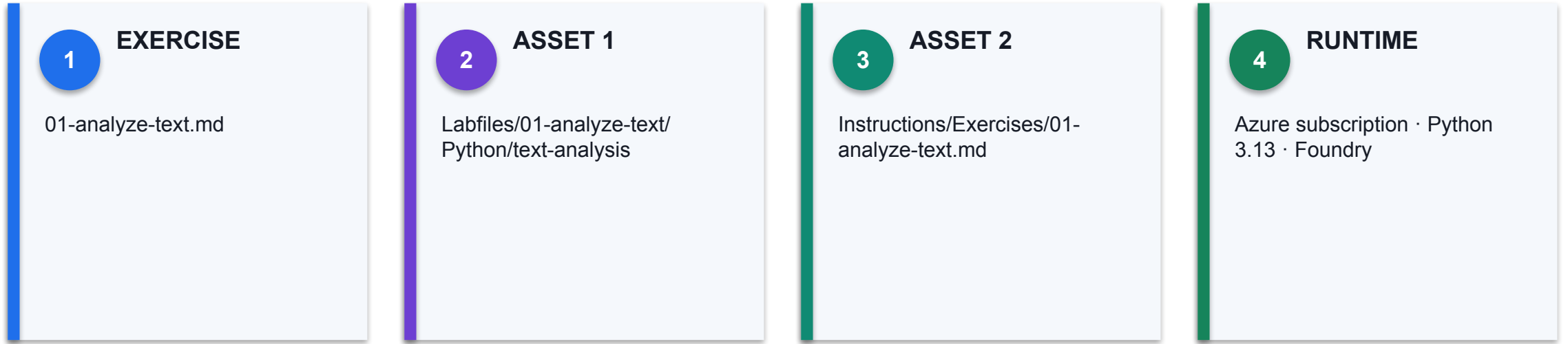
The official exercise completes with its expected output; a second learner can reproduce the result from the source instructions and the recorded environment.

Complete click-by-click procedure: [Learner Guide · Lab 07 · labs/lab-07-analyze-text/README.md](#)

Official lab 07 · pinned source

Contract	Exact value / field	Engineering consequence
Repository	mslearn-ai-language	MIT source snapshot
Revision	cea991780248	SHA-256 per file in source manifest
Exercise	01-analyze-text.md	Unchanged instruction file
Domain	Text analysis	AI-103 study guide

Official lab 07 · input boundaries



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Official lab 07 · implementation 1

```
# Get PII
pii_result =
ai_client.recognize_pii_entities(documents=[text])[0]
pii_entities = pii_result.entities
if len(pii_entities) > 0:
    print("\nPII Entities")
    for pii_entity in pii_entities:
        print('\t{} ({}).format(pii_entity.text,
pii_entity.category))
        print("Redacted Text:\n
{}".format(pii_result.redacted_text))
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 07 · implementation 2

```
# Get entities
entities =
ai_client.recognize_entities(documents=[text])
[0].entities
if len(entities) > 0:
    print("\nEntities")
    for entity in entities:
        print('\t{} ({}).format(entity.text,
entity.category))
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 07 · implementation 3

```
# Get language
detectedLanguage =
ai_client.detect_language(documents=[text])[0]
print('\nLanguage:
{}'.format(detectedLanguage.primary_language.name))
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

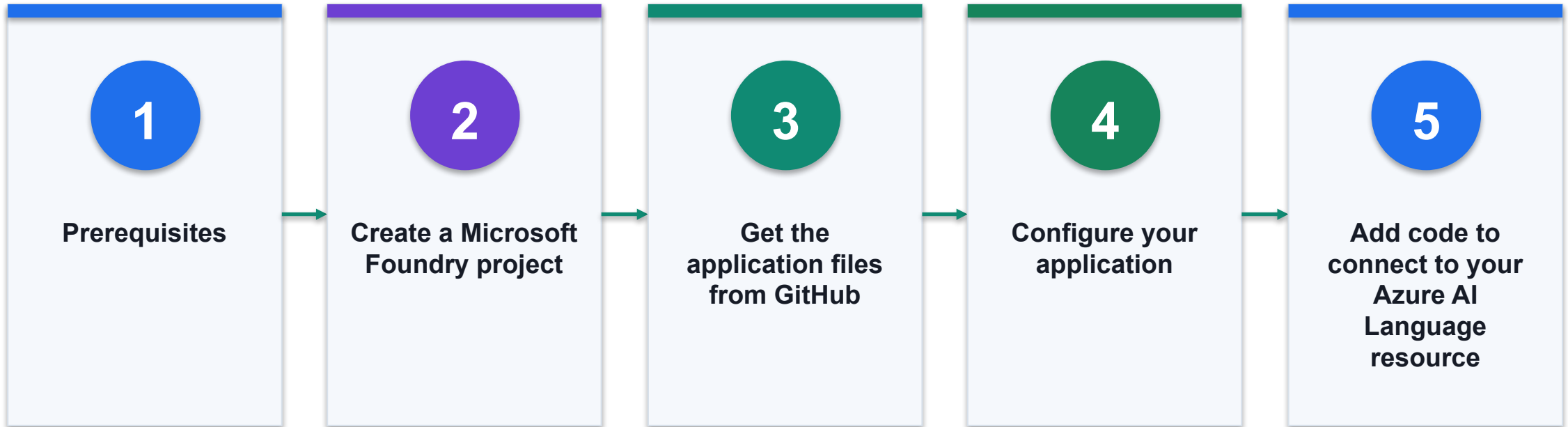
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 07 · mechanism phases



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Official lab 07 · source assets

Contract	Exact value / field	Engineering consequence
Asset 1	Labfiles/01-analyze-text/Python/text-analysis	Exact path in pinned source
Asset 2	Instructions/Exercises/01-analyze-text.md	Exact path in pinned source
Asset 3	LICENSE	Exact path in pinned source
Asset 4	README.md	Exact path in pinned source

Official lab 07 · evidence record

Contract	Exact value / field	Engineering consequence
Input	resource / model / source version	Capture before run
Output	result / request ID / trace	Compare to exercise expectation
Metric	quality / latency / cost	State unit and sample
Control	failure / limit / next action	Explain product implication

Lab 07: acceptance evidence

PASS WHEN

Complete the pinned Microsoft Learning exercise [official/mslearn-ai-language/Instructions/Exercises/01-analyze-text.md](#); capture its expected output, relevant version/endpoint, one measured result, and an explained failure or limitation.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

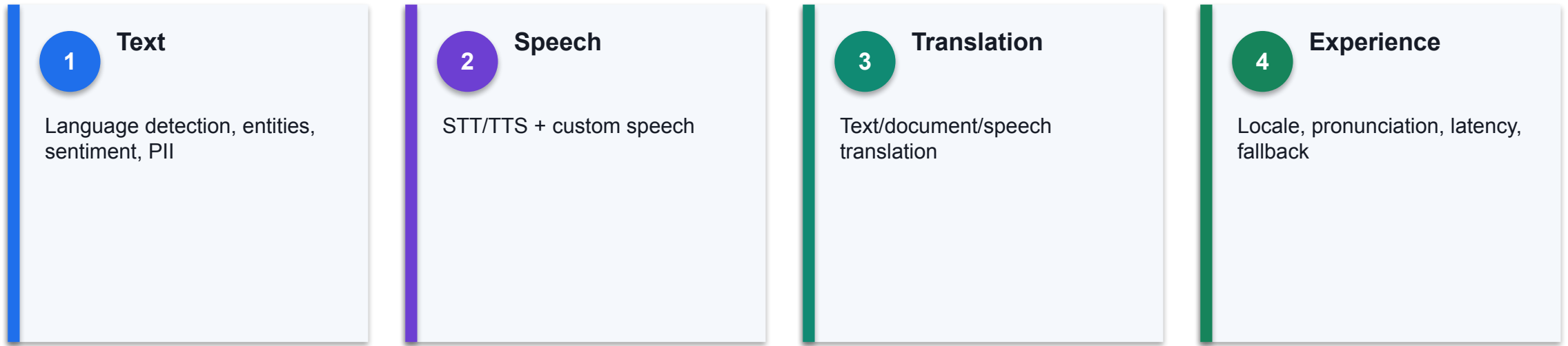
TOPIC 08

08

Develop a text analysis agent

TEXT ANALYSIS · AI-103

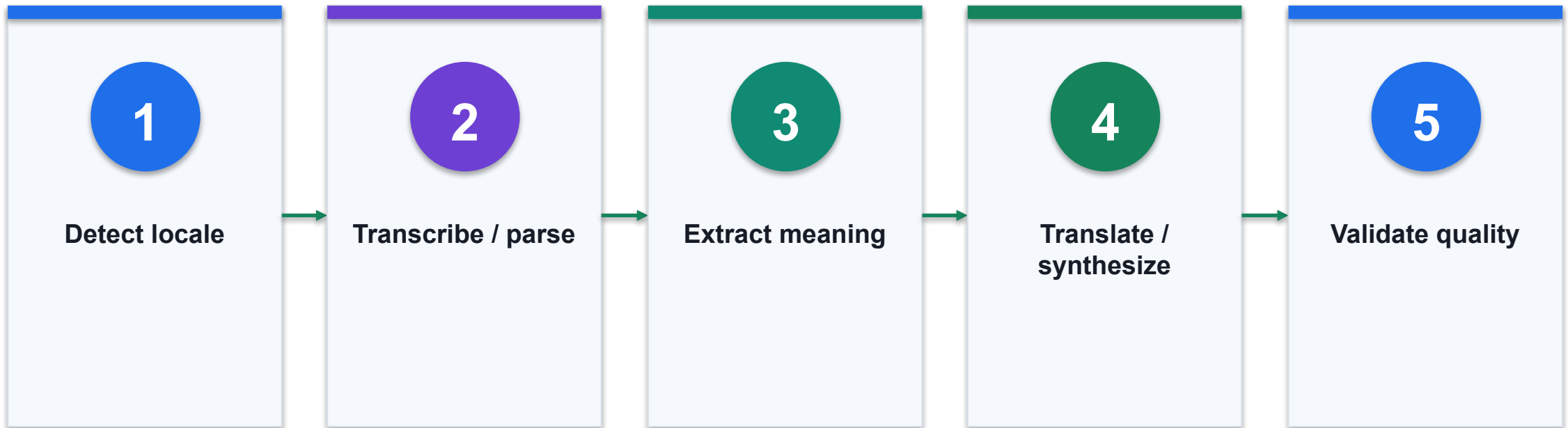
Develop a text analysis agent · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Develop a text analysis agent · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Develop a text analysis agent · exact contract

Contract	Exact value / field	Engineering consequence
language	en-SG / zh-Hans / auto	Affects model and voice
offset + length	Unicode text span	Preserve source mapping
SSML	voice, prosody, phoneme, break	Controlled speech output
recognition result	text + reason + duration	Handle NoMatch/Canceled

Develop a text analysis agent · implementation excerpt

```
<speak version="1.0" xml:lang="en-SG">
  <voice name="en-SG-LunaNeural">
    Your request is <prosody
rate="-5%">approved</prosody>.
  </voice>
</speak>
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

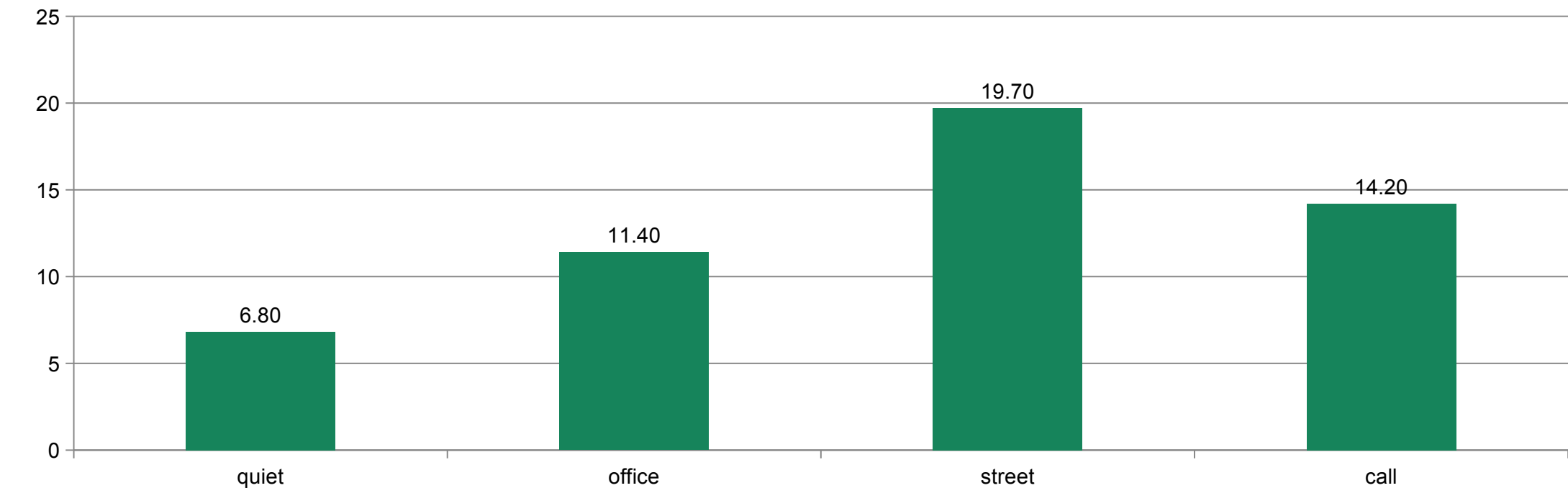
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

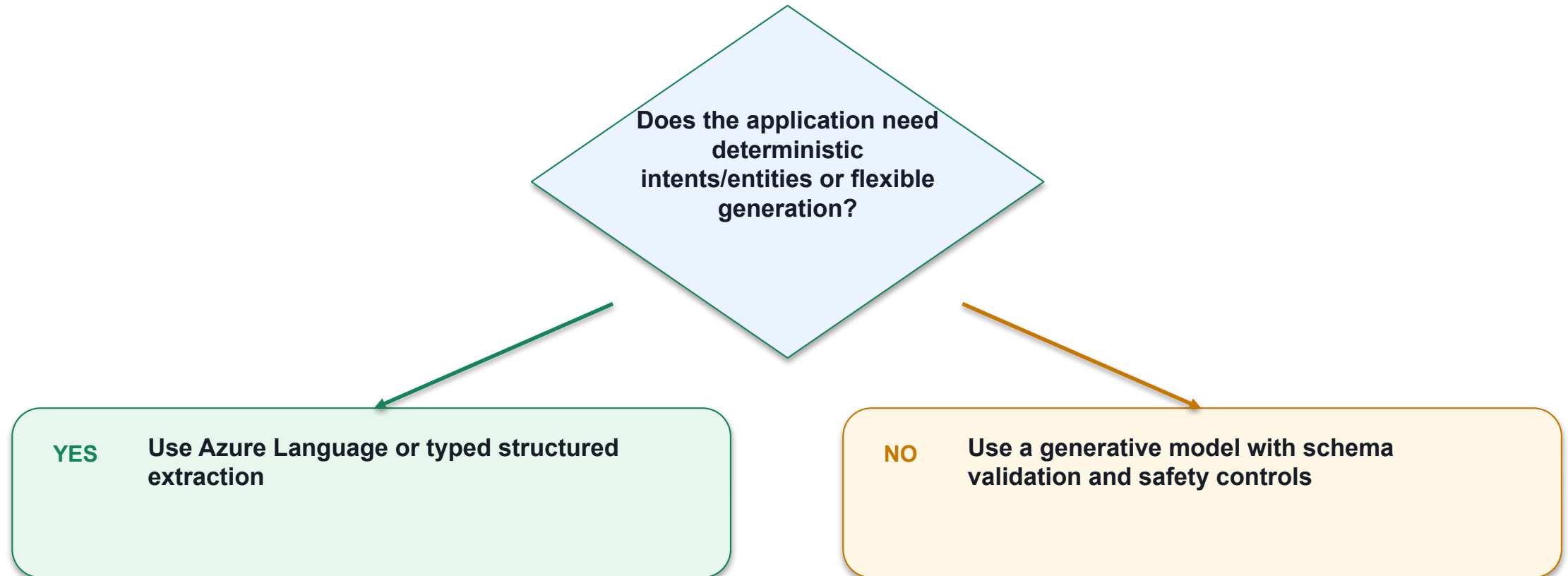
Handle auth, quota, timeout, and invalid-state paths.

Develop a text analysis agent · engineering metric



WHAT THE DATA SHOWS Indicative word error rate (%): acoustic context changes recognition quality. Test the real channel and vocabulary; do not accept a studio-only benchmark.

Develop a text analysis agent · decision rule



Develop a text analysis agent · failure and control

 NoMatch LIKELY CAUSE Audio/language mismatch CONTROL / RECOVERY Inspect reason and locale	 PII leak LIKELY CAUSE Transcript retained CONTROL / RECOVERY Redact before logging	 Bad pronunciation LIKELY CAUSE Voice/phoneme mismatch CONTROL / RECOVERY Use SSML phoneme/lexicon	 Translation drift LIKELY CAUSE Domain terms mistranslated CONTROL / RECOVERY Glossary/custom model + review
--	---	--	--

A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 08: Develop a text analysis agent

Mechanism focus: Develop a text analysis agent



YOU WILL PRODUCE

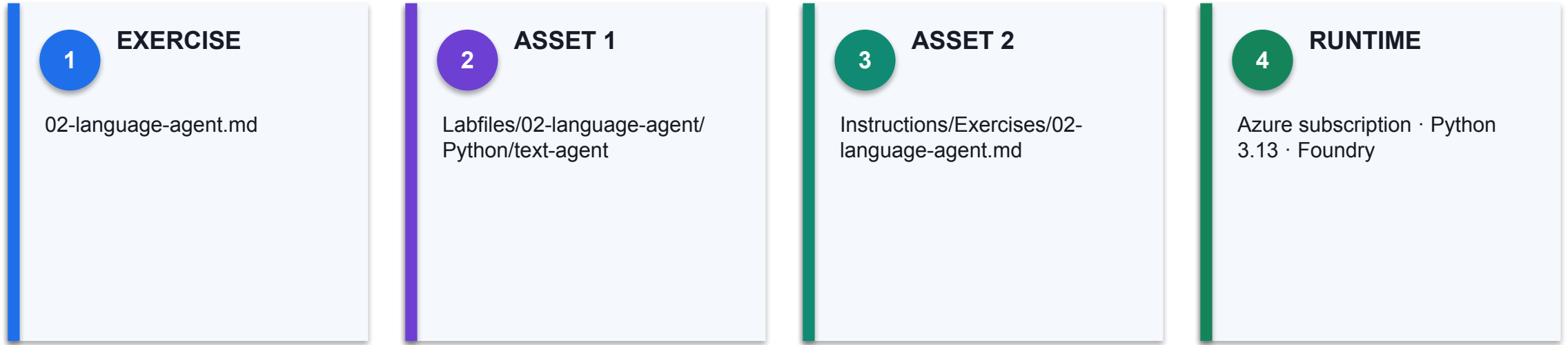
The official exercise completes with its expected output; a second learner can reproduce the result from the source instructions and the recorded environment.

Complete click-by-click procedure: [Learner Guide · Lab 08 · labs/lab-08-develop-a-text-analysis-agent/README.md](#)

Official lab 08 · pinned source

Contract	Exact value / field	Engineering consequence
Repository	mslearn-ai-language	MIT source snapshot
Revision	cea991780248	SHA-256 per file in source manifest
Exercise	02-language-agent.md	Unchanged instruction file
Domain	Text analysis	AI-103 study guide

Official lab 08 · input boundaries



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Official lab 08 · implementation 1

```
# Use the agent to get a response
prompt = input("User prompt: ")
response = openai_client.responses.create(
    input=[{"role": "user", "content": prompt}],
    extra_body={"agent_reference": {"name":
agent_name, "type": "agent_reference"}},
)

print(f"{agent_name}: {response.output_text}")
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 08 · implementation 2

```
# Get project client
project_client = AIProjectClient(
    endpoint=foundation_endpoint,
    credential=DefaultAzureCredential(),
)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 08 · implementation 3

```
# import namespaces
from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

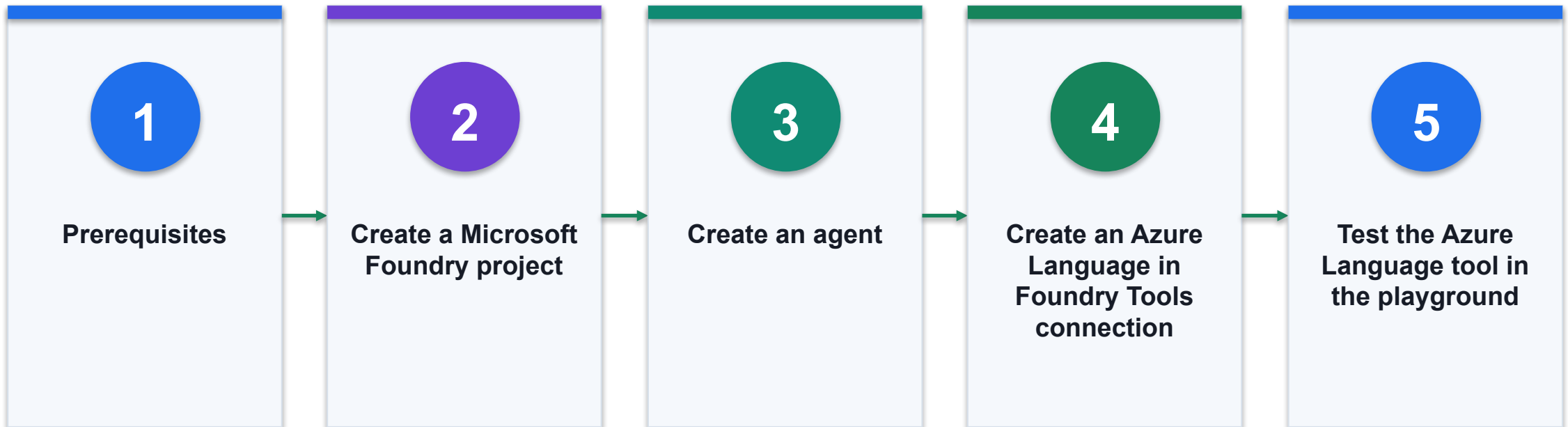
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 08 · mechanism phases



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Official lab 08 · source assets

Contract	Exact value / field	Engineering consequence
Asset 1	Labfiles/02-language-agent/Python/text-agent	Exact path in pinned source
Asset 2	Instructions/Exercises/02-language-agent.md	Exact path in pinned source
Asset 3	LICENSE	Exact path in pinned source
Asset 4	README.md	Exact path in pinned source

Official lab 08 · evidence record

Contract	Exact value / field	Engineering consequence
Input	resource / model / source version	Capture before run
Output	result / request ID / trace	Compare to exercise expectation
Metric	quality / latency / cost	State unit and sample
Control	failure / limit / next action	Explain product implication

Lab 08: acceptance evidence

PASS WHEN

Complete the pinned Microsoft Learning exercise [official/mslearn-ai-language/Instructions/Exercises/02-language-agent.md](#); capture its expected output, relevant version/endpoint, one measured result, and an explained failure or limitation.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

TOPIC 09

09

Extract multimodal information

INFORMATION EXTRACTION · AI-103

Extract multimodal information



Input

Document/image/audio/video + content type

Analyzer

Prebuilt/custom/composed/Content Understanding

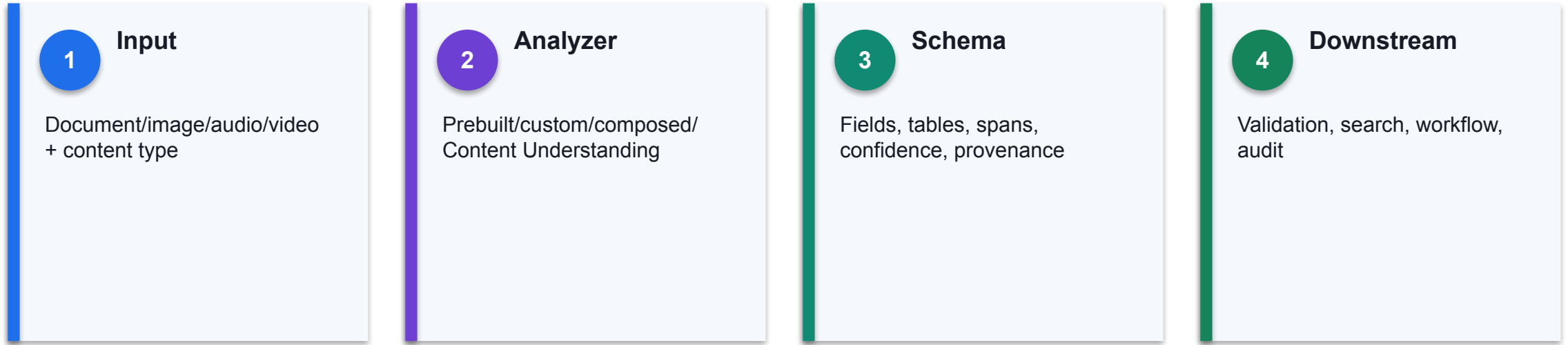
Schema

Fields, tables, spans, confidence, provenance

Downstream

Validation, search, workflow, audit

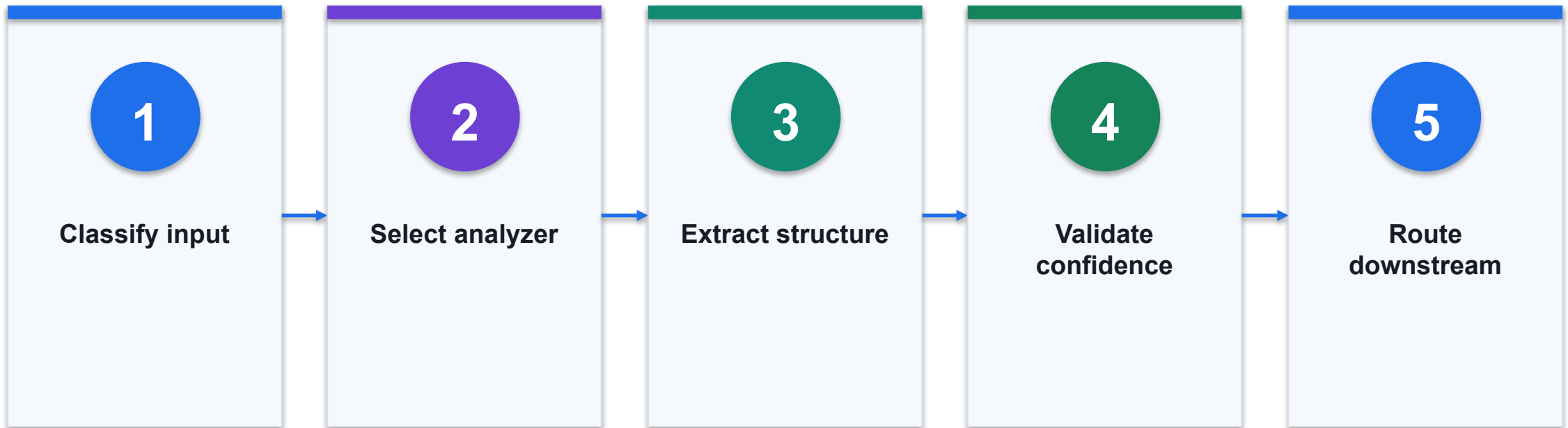
Extract multimodal information · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Extract multimodal information · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Extract multimodal information · exact contract

Contract	Exact value / field	Engineering consequence
modelId / analyzerId	prebuilt or versioned custom	Release identity
pages / spans	offset + length + polygon	Trace extracted value to source
confidence	field-level 0–1	Drive review threshold
operation-location	async result URL	Poll until succeeded/failed

Extract multimodal information · implementation excerpt

```
poller = client.begin_analyze_document(  
    model_id="prebuilt-invoice",  
    body=document_bytes,  
    content_type="application/pdf")  
result = poller.result()  
for doc in result.documents:  
    print(doc.fields["InvoiceTotal"].value_currency)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

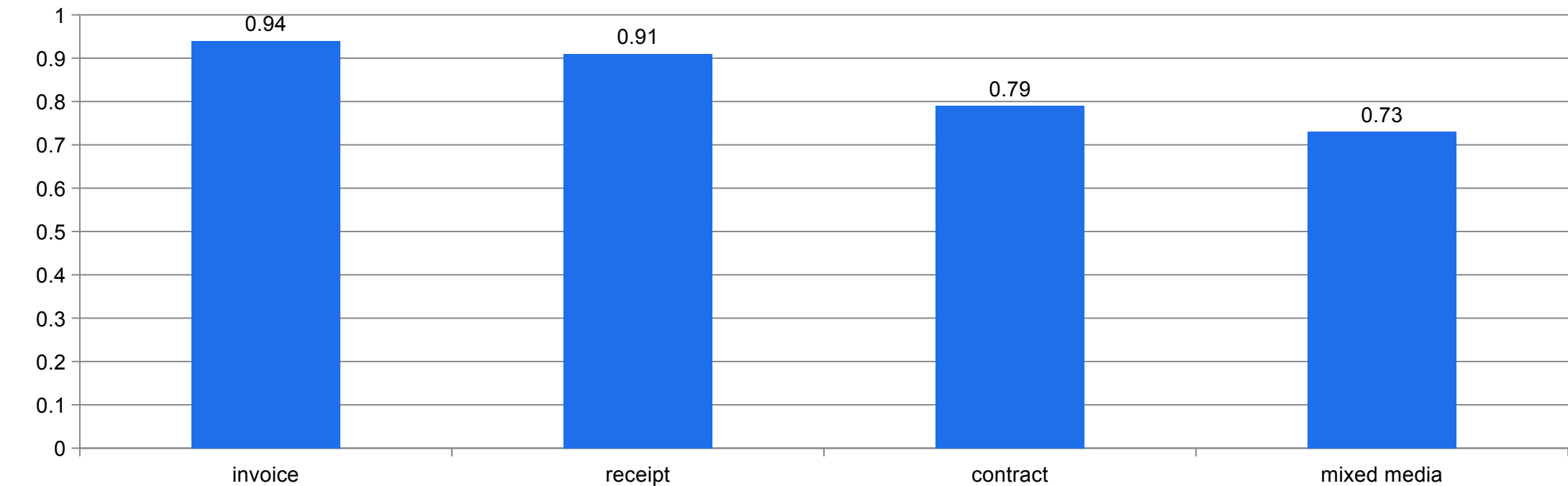
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

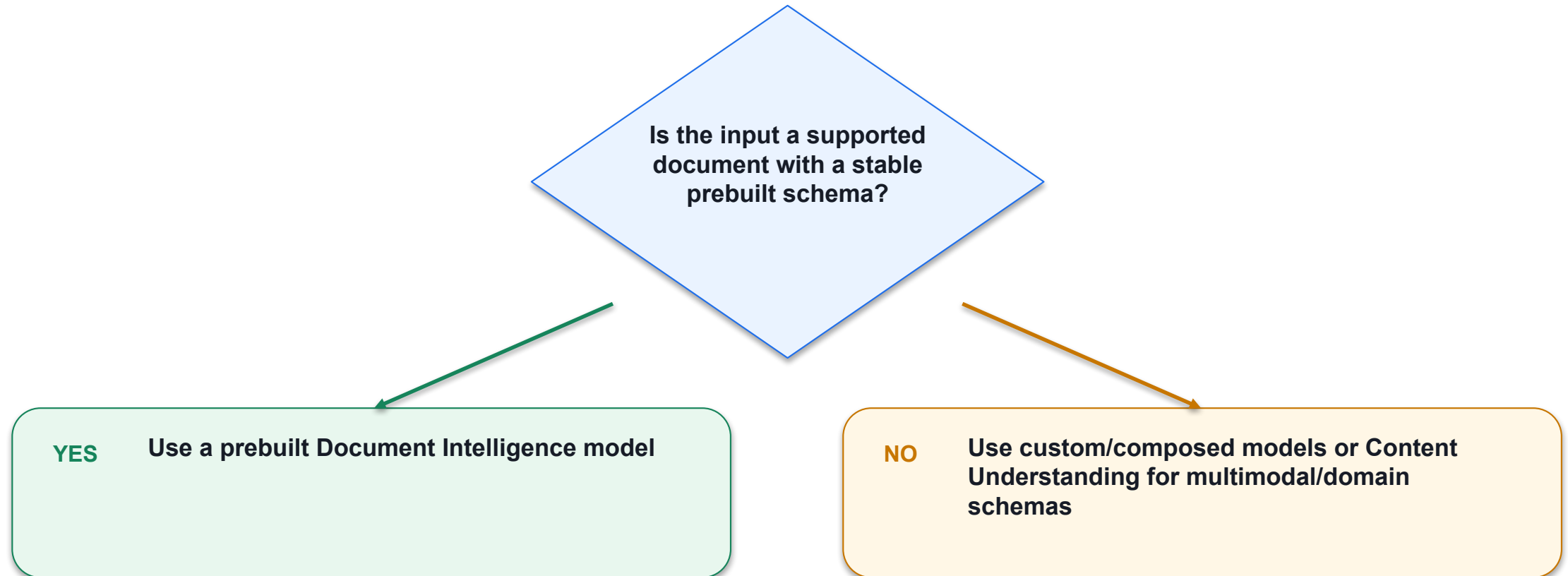
Handle auth, quota, timeout, and invalid-state paths.

Extract multimodal information - engineering metric



WHAT THE DATA SHOWS Indicative field F1: prebuilt models excel on supported formats; custom/composed models or Content Understanding address domain-specific and multimodal schemas.

Extract multimodal information · decision rule



Extract multimodal information - failure and control

 Wrong model LIKELY CAUSE Document type misclassified CONTROL / RECOVERY Classify first or use composed model	 Low confidence LIKELY CAUSE Layout/domain drift CONTROL / RECOVERY Human review + retraining sample	 Async timeout LIKELY CAUSE Polling/size/page limits CONTROL / RECOVERY Honor operation status and limits	 Lost provenance LIKELY CAUSE Value stored without span/page CONTROL / RECOVERY Persist source URI + region/span
---	--	---	--

A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 09: Extract multimodal information

Mechanism focus: Extract multimodal information



YOU WILL PRODUCE

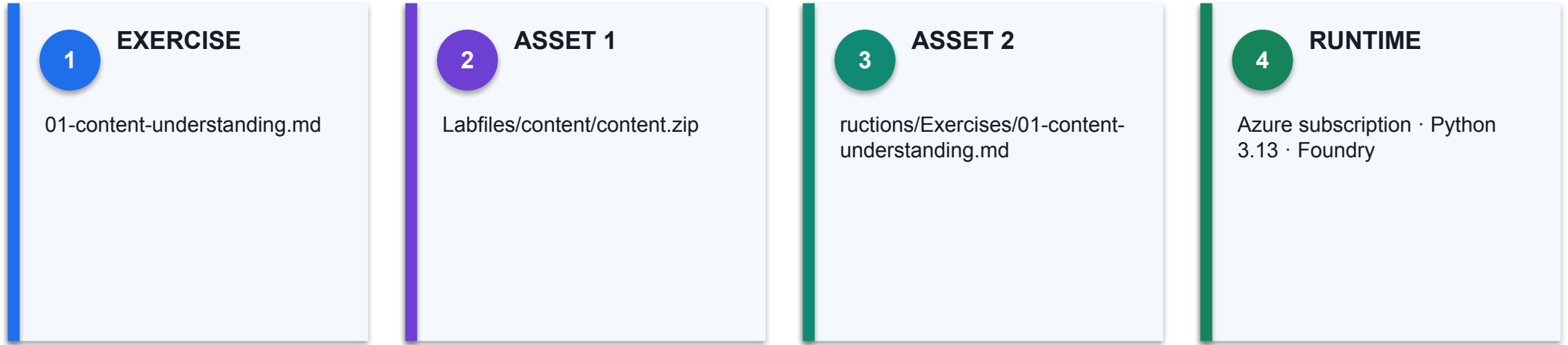
The official exercise completes with its expected output; a second learner can reproduce the result from the source instructions and the recorded environment.

Complete click-by-click procedure: [Learner Guide · Lab 09 · labs/lab-09-extract-multimodal-information/README.md](#)

Official lab 09 · pinned source

Contract	Exact value / field	Engineering consequence
Repository	mslearn-ai-information-extraction	MIT source snapshot
Revision	a79b11a76caf	SHA-256 per file in source manifest
Exercise	01-content-understanding.md	Unchanged instruction file
Domain	Information extraction	AI-103 study guide

Official lab 09 · input boundaries



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Official lab 09 · implementation 1

```
poller = client.begin_analyze_document(  
    model_id="prebuilt-invoice",  
    body=document_bytes,  
    content_type="application/pdf")  
result = poller.result()  
for doc in result.documents:  
    print(doc.fields["InvoiceTotal"].value_currency)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 09 · implementation 2

```
modelId / analyzerId: prebuilt or versioned custom #  
Release identity  
pages / spans: offset + length + polygon # Trace  
extracted value to source  
confidence: field-level 0–1 # Drive review threshold  
operation-location: async result URL # Poll until  
succeeded/failed
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 09 · implementation 3

Input: Document/image/audio/video + content type
Analyzer: Prebuilt/custom/composed/Content
Understanding
Schema: Fields, tables, spans, confidence, provenance
Downstream: Validation, search, workflow, audit

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

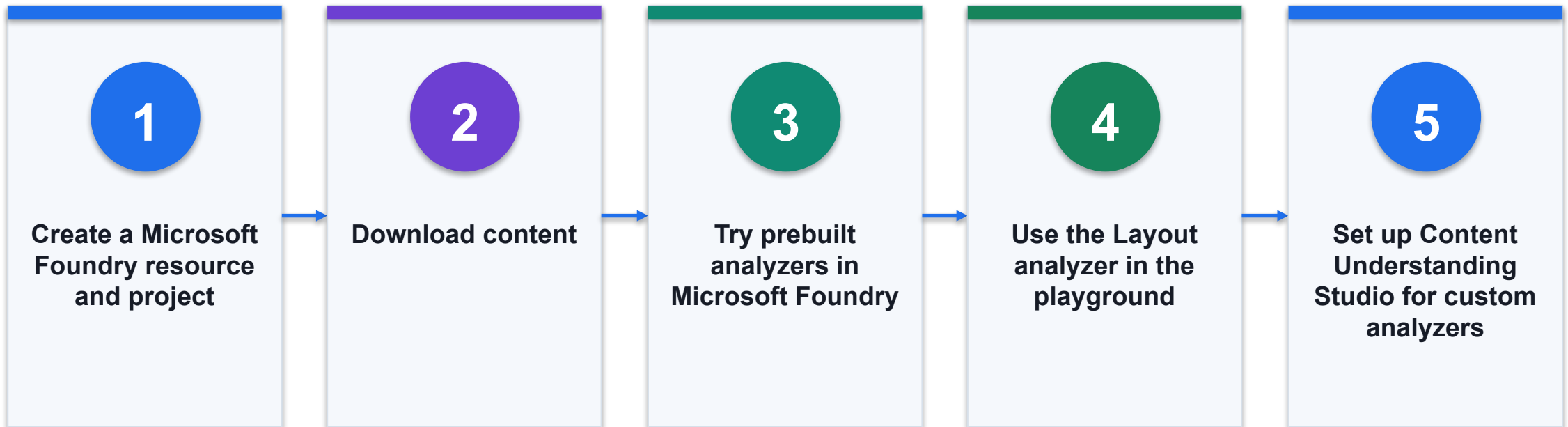
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 09 · mechanism phases



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Official lab 09 · source assets

Contract	Exact value / field	Engineering consequence
Asset 1	Labfiles/content/content.zip	Exact path in pinned source
Asset 2	Instructions/Exercises/01-content-understanding.md	Exact path in pinned source
Asset 3	LICENSE	Exact path in pinned source
Asset 4	README.md	Exact path in pinned source

Official lab 09 · evidence record

Contract	Exact value / field	Engineering consequence
Input	resource / model / source version	Capture before run
Output	result / request ID / trace	Compare to exercise expectation
Metric	quality / latency / cost	State unit and sample
Control	failure / limit / next action	Explain product implication

Lab 09: acceptance evidence

PASS WHEN

Complete the pinned Microsoft Learning exercise [official/mslearn-ai-information-extraction/Instructions/Exercises/01-content-understanding.md](#); capture its expected output, relevant version/endpoint, one measured result, and an explained failure or limitation.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

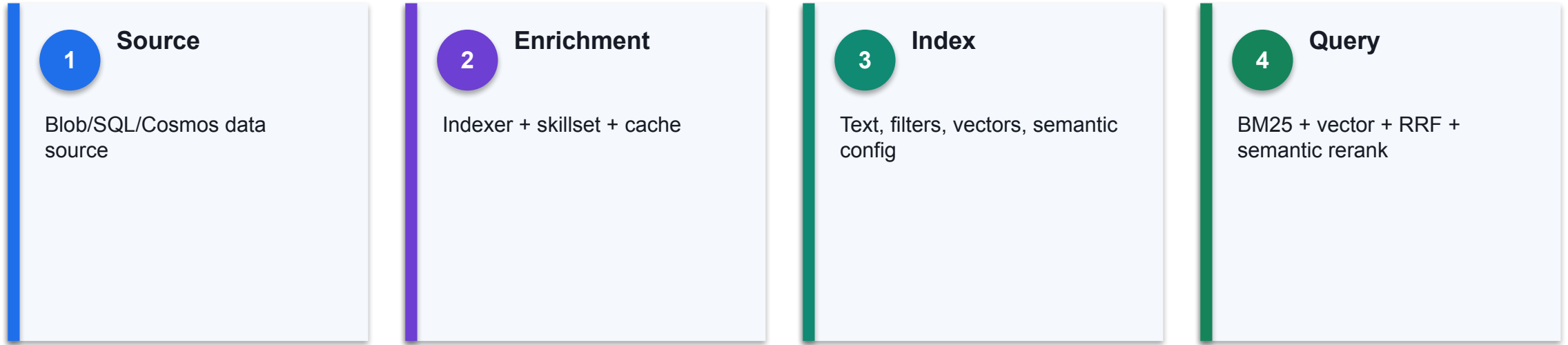
TOPIC 10

10

Build a knowledge mining solution

INFORMATION EXTRACTION · AI-103

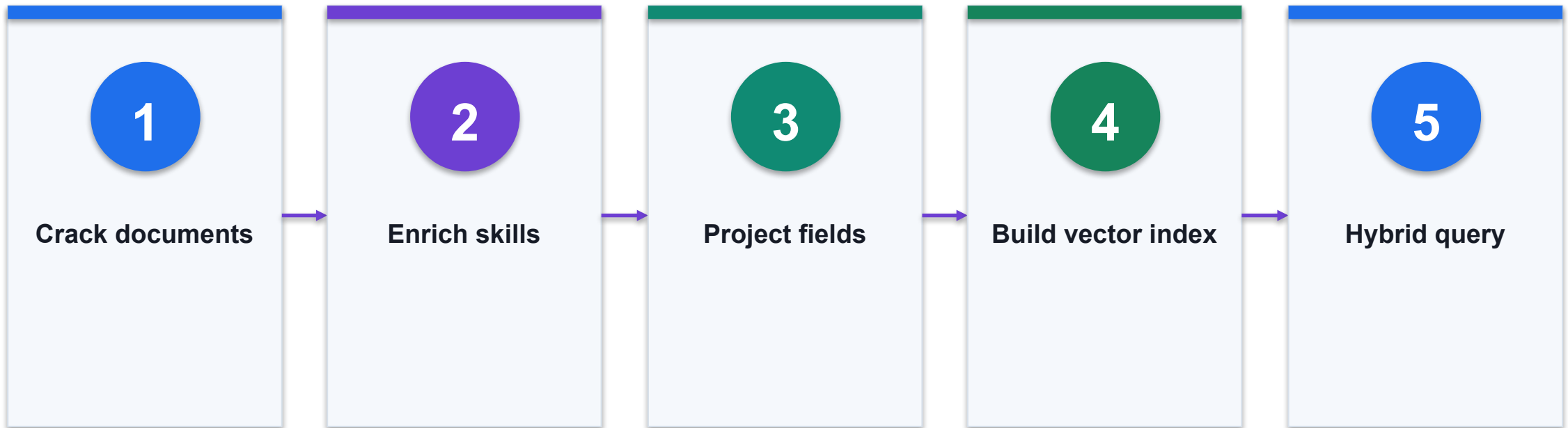
Build a knowledge mining solution · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Build a knowledge mining solution · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Build a knowledge mining solution · exact contract

Contract	Exact value / field	Engineering consequence
key	Edm.String, key=true	Unique document/chunk ID
vector	Collection(Edm.Single)	Dimensions + HNSW profile
search/filter/select	query contract	Reduce exposure and payload
scores	@search.score / rerankerScore	Different scales; do not compare directly

Build a knowledge mining solution · implementation excerpt

```
{
  "search": "maintenance policy",
  "vectorQueries": [{"kind": "text", "text":
"maintenance policy",
+      "fields": "content_vector"}],
  "queryType": "semantic",
  "filter": "tenant_id eq 'T-42'",
  "top": 5
}
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

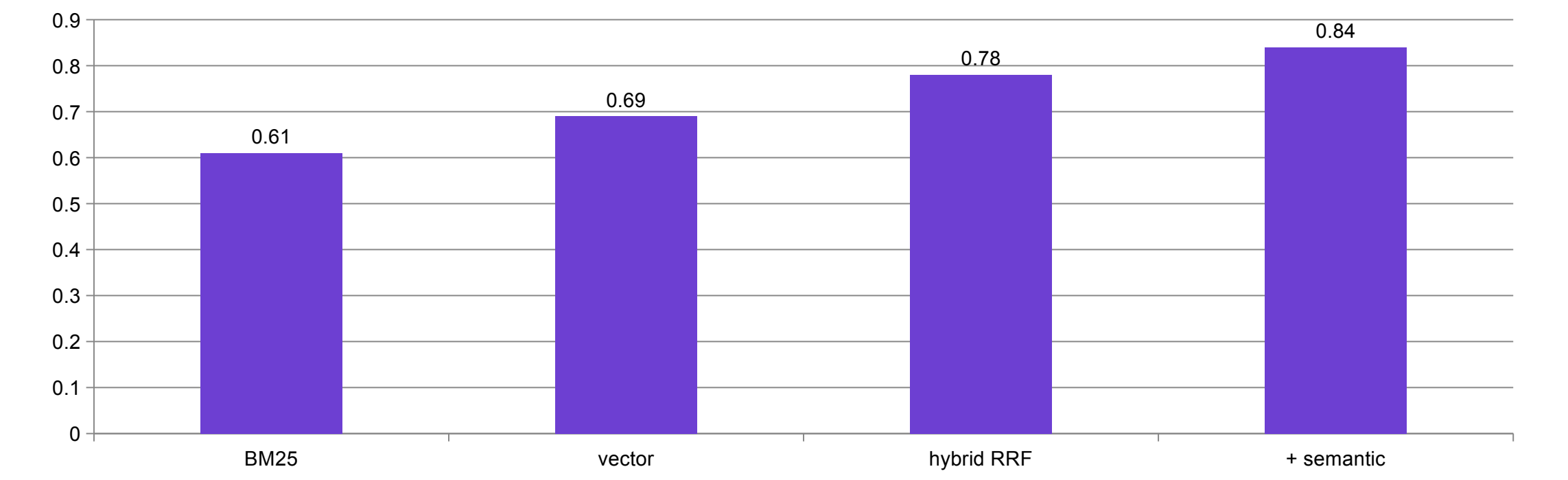
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

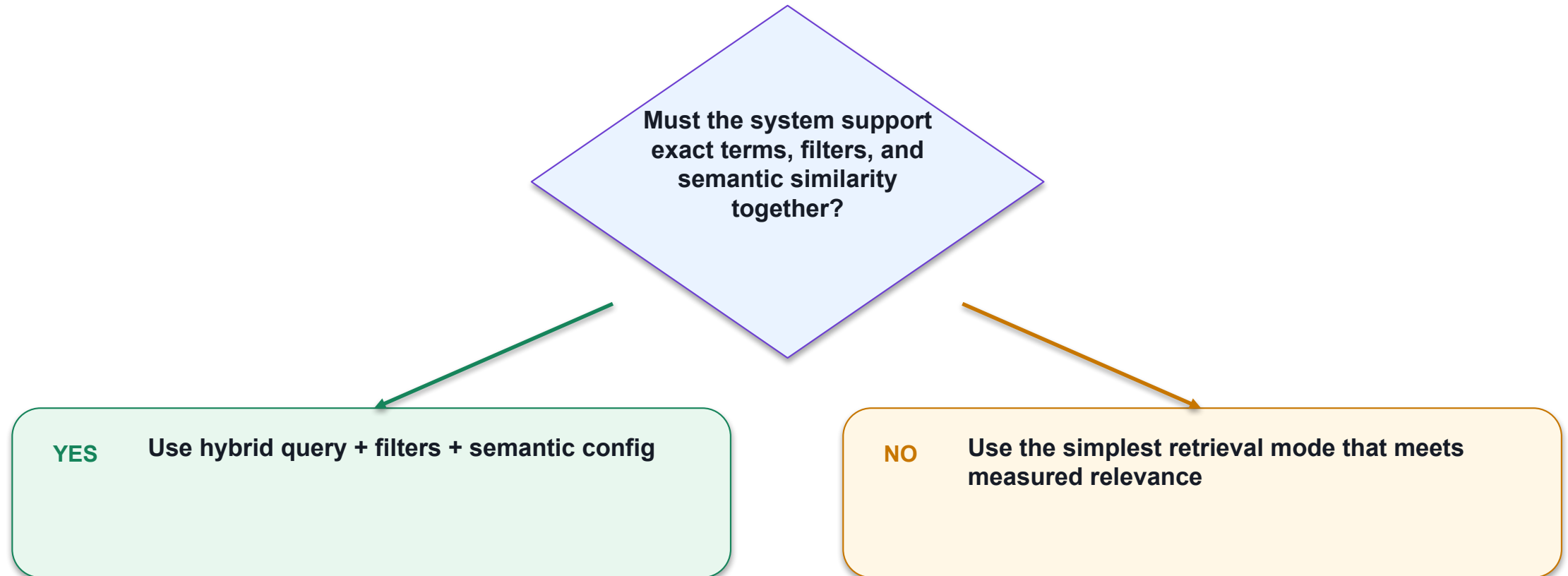
Handle auth, quota, timeout, and invalid-state paths.

Build a knowledge mining solution · engineering metric



WHAT THE DATA SHOWS Indicative nDCG@5: hybrid retrieval often improves coverage, while semantic reranking improves ordering. Validate with your own judged query set.

Build a knowledge mining solution · decision rule



Build a knowledge mining solution · failure and control

 Indexer error LIKELY CAUSE Skill input path missing CONTROL / RECOVERY Inspect enrichment tree and field mappings	 Empty vector LIKELY CAUSE Embedding skill/vectorizer failed CONTROL / RECOVERY Check endpoint, dimensions, projection	 Poor relevance LIKELY CAUSE No judged query set CONTROL / RECOVERY Create labels and compare retrieval modes	 Data leak LIKELY CAUSE filterable security field absent CONTROL / RECOVERY Enforce tenant/ACL filter before results
--	--	---	--

A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 10: Build a knowledge mining solution

Mechanism focus: Build a knowledge mining solution



YOU WILL PRODUCE

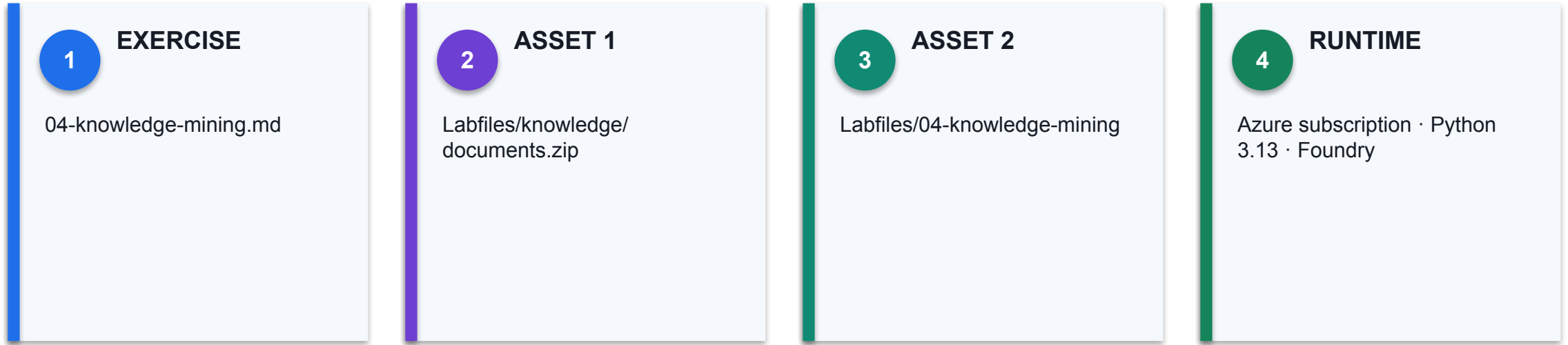
The official exercise completes with its expected output; a second learner can reproduce the result from the source instructions and the recorded environment.

Complete click-by-click procedure: [Learner Guide · Lab 10 · labs/lab-10-build-a-knowledge-mining-solution/README.md](#)

Official lab 10 · pinned source

Contract	Exact value / field	Engineering consequence
Repository	mslearn-ai-information-extraction	MIT source snapshot
Revision	a79b11a76caf	SHA-256 per file in source manifest
Exercise	04-knowledge-mining.md	Unchanged instruction file
Domain	Information extraction	AI-103 study guide

Official lab 10 · input boundaries



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Official lab 10 · implementation 1

```
{  
  "search": "New York",  
  "count": true,  
  "select": "title,keyPhrases",  
  "filter": "metadata_storage_size lt 380000"  
}
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 10 · implementation 2

```
{  
  "search": "New York",  
  "count": true,  
  "select": "title,keyPhrases"  
}
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 10 · implementation 3

```
python -m venv labenv
labenv\Scripts\activate
pip install -r requirements.txt
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

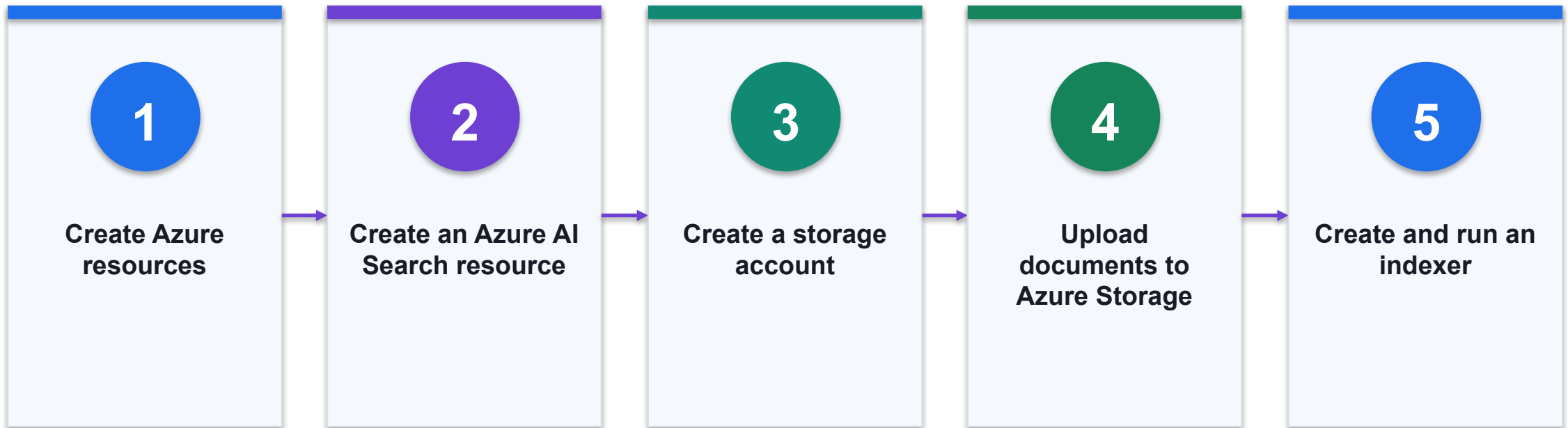
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

Official lab 10 · mechanism phases



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Official lab 10 · source assets

Contract	Exact value / field	Engineering consequence
Asset 1	Labfiles/knowledge/documents.zip	Exact path in pinned source
Asset 2	Labfiles/04-knowledge-mining	Exact path in pinned source
Asset 3	Instructions/Exercises/04-knowledge-mining.md	Exact path in pinned source
Asset 4	LICENSE	Exact path in pinned source

Official lab 10 · evidence record

Contract	Exact value / field	Engineering consequence
Input	resource / model / source version	Capture before run
Output	result / request ID / trace	Compare to exercise expectation
Metric	quality / latency / cost	State unit and sample
Control	failure / limit / next action	Explain product implication

Lab 10: acceptance evidence

PASS WHEN

Complete the pinned Microsoft Learning exercise [official/mslearn-ai-information-extraction/Instructions/Exercises/04-knowledge-mining.md](#); capture its expected output, relevant version/endpoint, one measured result, and an explained failure or limitation.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

TOPIC 00

00

Assessment and course close

Evidence submission · assessment flow · support

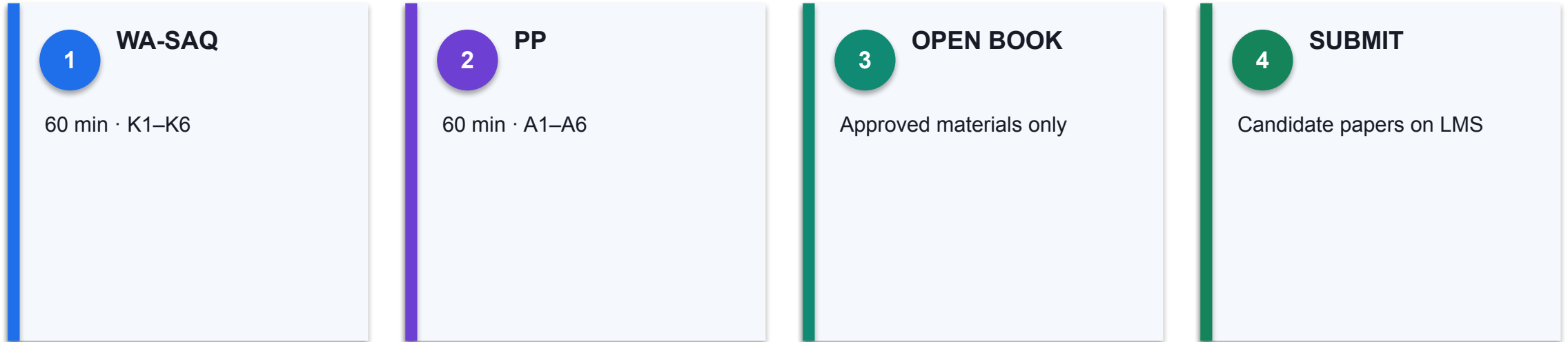
Course enquiries and support

Contact Tertiary Courses for current course details, schedules, and registration

[Tertiary Courses · Course Enquiries](#)

OPEN RESOURCE

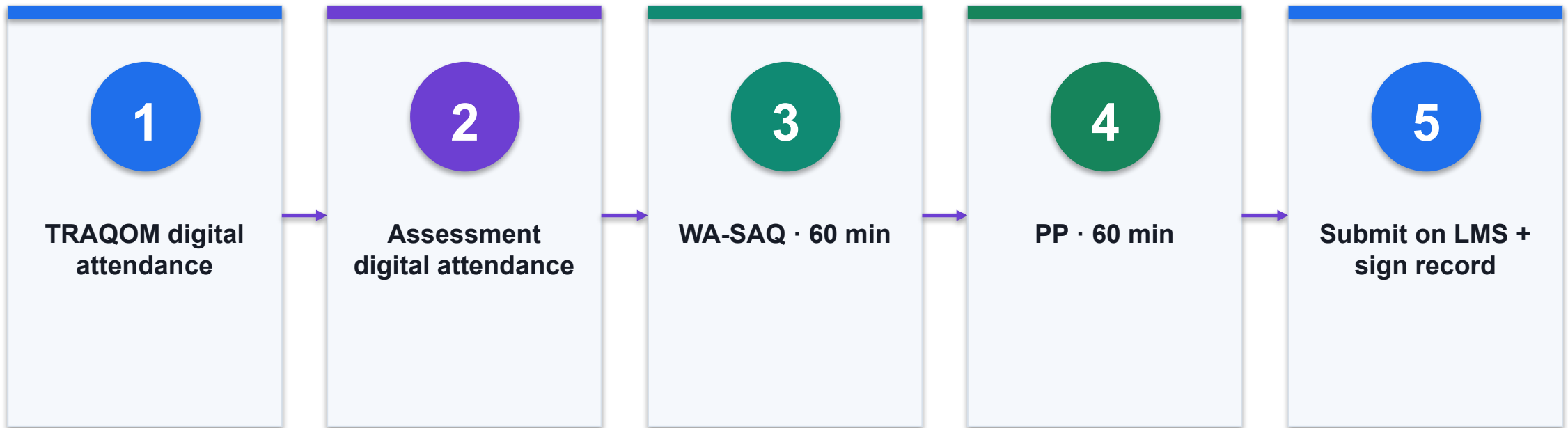
Assessment



ARCHITECTURE RULE

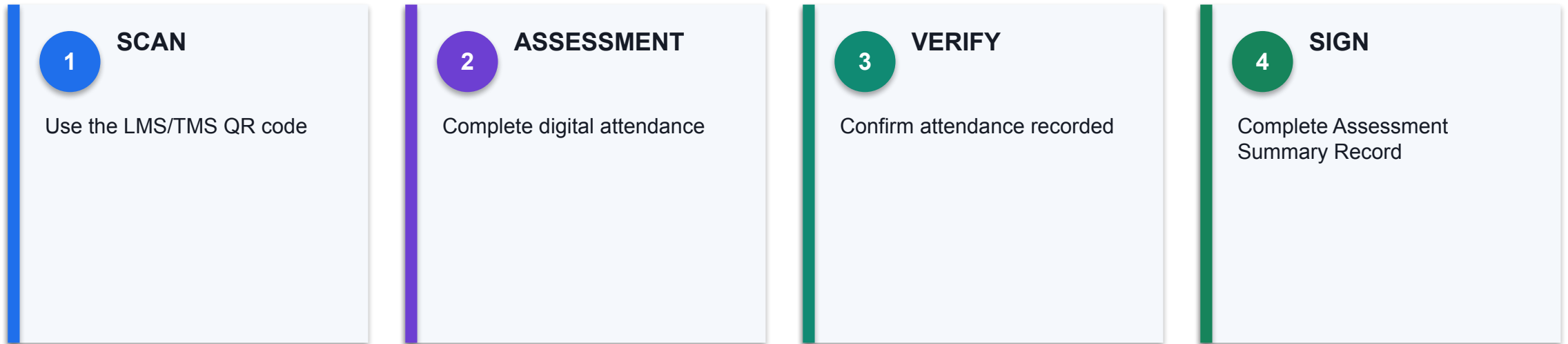
Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Assessment Flow



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Digital Attendance (Mandatory)



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

TOPIC 00

00

Thank You

You can now explain, test, and operate an end-to-end Azure AI solution